Repair Manual

911 Carrera
(993)

Volume VI:
Air conditioning
Vehicle electrics

Dr. Ing. h.c. F. Porsche Aktiengesellschaft

Preface

Structure

The "Technical Literture" for the "911 Carrera (993)" model is basically structured as before, i.e. the structure follows the familiar repair groups.

A new feature is that the structure includes the main groups **0 to 9** and the main group **D**.

Main groups:	0	Complete vehicle – General
	1	Engine
	2	Fuel, exhaust, engine electrical system
	3	Transmission
	4	Chassis
	5	Body
	6	Body equipment, outside
	7	Body equipment, interior
	8	Air conditioning
	9	Electrical system
	D	Diagnosis

Layout

The layout in the below items remains unchanged throughout the repair manual

1. Table of tightening torques
2. Special tools required
3. Exploded views
4. Legends for the exploded views
5. Assembly notes / use of special tools

As a new feature, however, the former item 6 (Repair group diagnosis) is no longer filed in the volume corresponding to the respective repair group. The **Diagnosis test plans / diagnosis procedures** have been combined in a **separate Diagnosis volume** broken down according to the main groups 0 to 9.

Another new feature is that the contents of the "Service Information Technik" are indicated in the Repair Manual. This brochure concentrates on a description of the design and function of components and of the new features introduced for a particular model year.

Service Number

All major repair procedures and repair descriptions are identified by a two- or four-digit **Service Number** completed by two additional digits to identify the work that corresponds to the first six digits of the working position number in the Working Times and Damage Catalog.

Example: 30 37 37 Dismantling and assembling clutch control shaft

Explanation: 30 37 37 50 (full working position number)

Repair group
here: Clutch, control

Component designation
here: Clutch control shaft

Activity
here: Dismantling and assembling

Index
here: Removed

Presentation in the various documents

30 37 37 50	Working position no. from **Working Times and Damage Catalog**, consisting of repair group, component designation, activity and index
30 37 37	Six-digit number in **Repair Manual**, consisting of repair group, component designation and activity
30 37	Service number in **Service Information**, consisting of repair group and component designation

Goal

The introduction of a service number in the "technical literature" is intended to facilitate standardization and positive identification to allow direct cross-referencing among the various documents. This is of particular importance with regard to the use of electronic media.

VI Air conditioning, Vehicle Electrics

The **Repair Manual of the 911 Carrera (993)** also includes the **911 Carrera 4 manual (993 four-wheel drive)**. The 911 Carrera (993) is the basic model covered by the repair operations described in this Manual. "911 Carrera (993)" is also indicated in the header of each page.

Descriptions of repair operations that deviate for the 911 Carrera 4 will be included after the respective 911 Carrera section. The repair descriptions of both models are separated by a cover page. All pages included after the cover page (separation sheet) have the "911 Carrera 4" heading. To facilitate distinction, **the page numbering will start with 100.**

Where different or additional repair procedures apply to the 911 Targa (993), these are given following the repair instructions for the 911 Carrera or 911 Carrera 4. The repair instructions for the 911 Targa (993) are separated by a title page. "**911 Targa (993)**" appears at the top of each page after in the title page (divider). In addition, **the page numbering starts with 200.**

Printed in Germany - 15, 1996

Printed in Germany - 16, 1996

Survey of contents of Service Information Technik '95

The Service Information gives a detailed description of the technical features of the new 911 Carrera.

8 Technical data of air conditioning system

Compressor model 10 PA 15 C

Refrigerant charge 840 g refrigerant **R 134a**

Refrigerant oil in compressor 140 ± 20 cm^3 ND 8

Tightening torques for refrigerant lines

Thread outer dia.	TPI	Tightening torques in Nm
5/8"	18 UNF	17 ± 3
3/4"	16 UNF	24 ± 4
7/8"	14 UNF	33 ± 4

Hex bolts at	Thread	Tightening torques in Nm
Expansion valve	M 5	6
Expansion valve	M 6	9
Compressor	M 8	28

Note

When fitting the refrigerant lines, apply some refrigerant oil to the bolt unions and the O-rings.

Dispose of the refrigerant oil as **hazardous waste.**

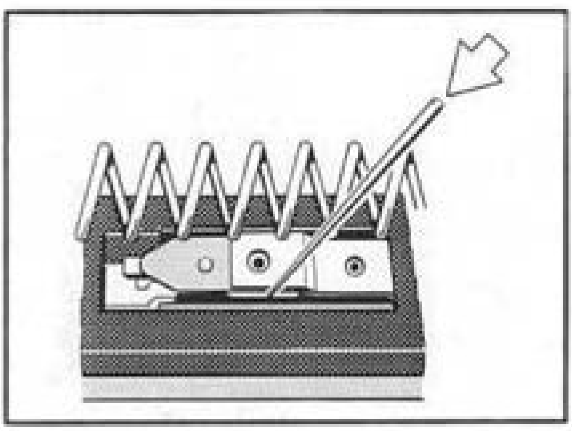
80 59 01 Checking ballast resistor of rear blower

Note

The ballast resistor is fitted with a restart lock-out device. This restart lockout may be by-passed if the 1st stage of the rear blower should fail (2nd stage of rear blower operates continuously).

1. Take out ballast resistor.

2. Press down spring plate with a needle or a similar tool. This causes the contact tongue to snap back into its initial position.



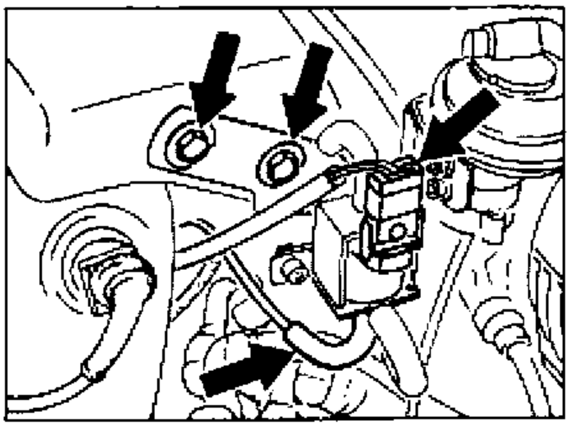
200-80

3. Replace the ballast resistor if the contacts are burnt.

80 18 19 Removing and installing rear heater blower

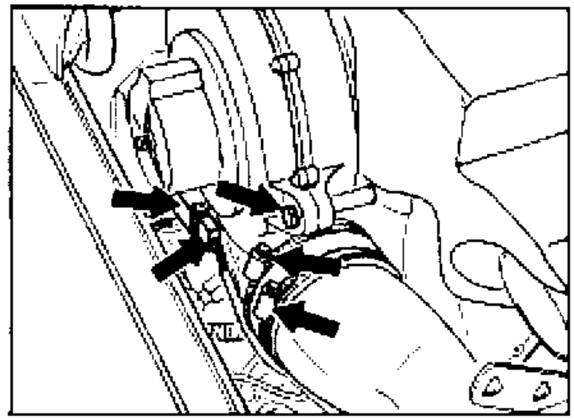
Removal

1. Remove intercooler. Pull vacuum hose and connector off the switchover valve. Undo fastening screws from fresh air housing and take out fresh air housing.



2027-10

2. Remove hose clamps and tie-wraps from heating air elbow and pull off rubber boot. Disconnect connector from rear heater blower and undo fastening screws (2 screws).



2073-80

3. Pull rear heater blower out of support in rearward direction and take out blower.

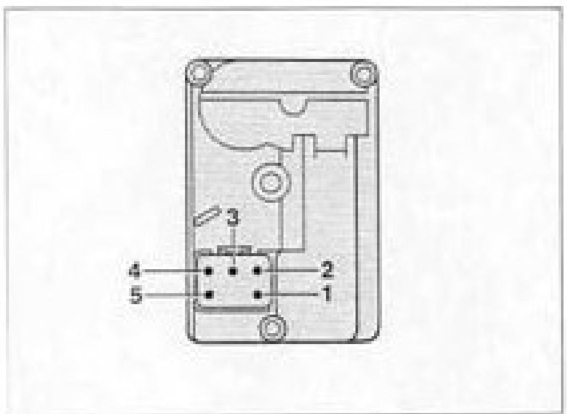
Installation

Check for correct seating of hose connections and hose clamps on heater air elbow.

85 16 15 Adjusting the flaps of the heater/ A/C unit

Adjusting the defroster center nozzle flap

1. Set motor to "Defroster closed" end position: Apply a voltage of 12 volts to pin 4 (positive) and pin 5 (negative) using two jumper leads until the motor is positioned in the end position.



256-87

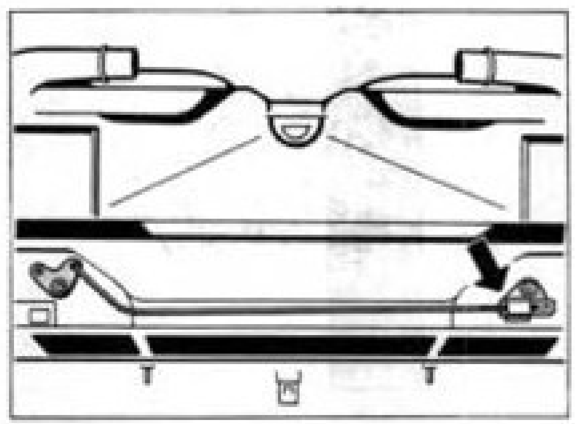
2. Press defroster center nozzle flap into "Defroster nozzle closed" position (upper outlet closed).

3. Assemble motor drive with flap link and lock assembly.

Adjusting the footwell flaps

1. Set motor to "Footwell flaps closed" end position: apply a voltage of 12 volts to pin 4 (negative) and pin 5 (positive) using two jumper leads until the motor is positioned in the end position. The drive lever now points towards the fresh air inlet.

2. Set both footwell flaps to closed position. Engage linkage with lever and locate with clamp.



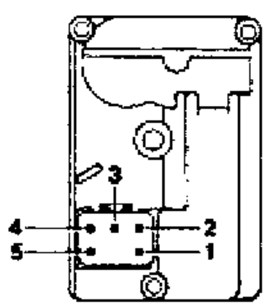
258-87

85 51 15 Adjusting the temperature mixing valves

Note

The temperature mixing vales are located below the instrument panel in the driver and passenger footwells.

1. Set engine to "temperature mixing valves closed" end position: Use two jump leads to apply 12 volts to pin 4 (positive) and pin 5 (negative) until the motor is in the end position.

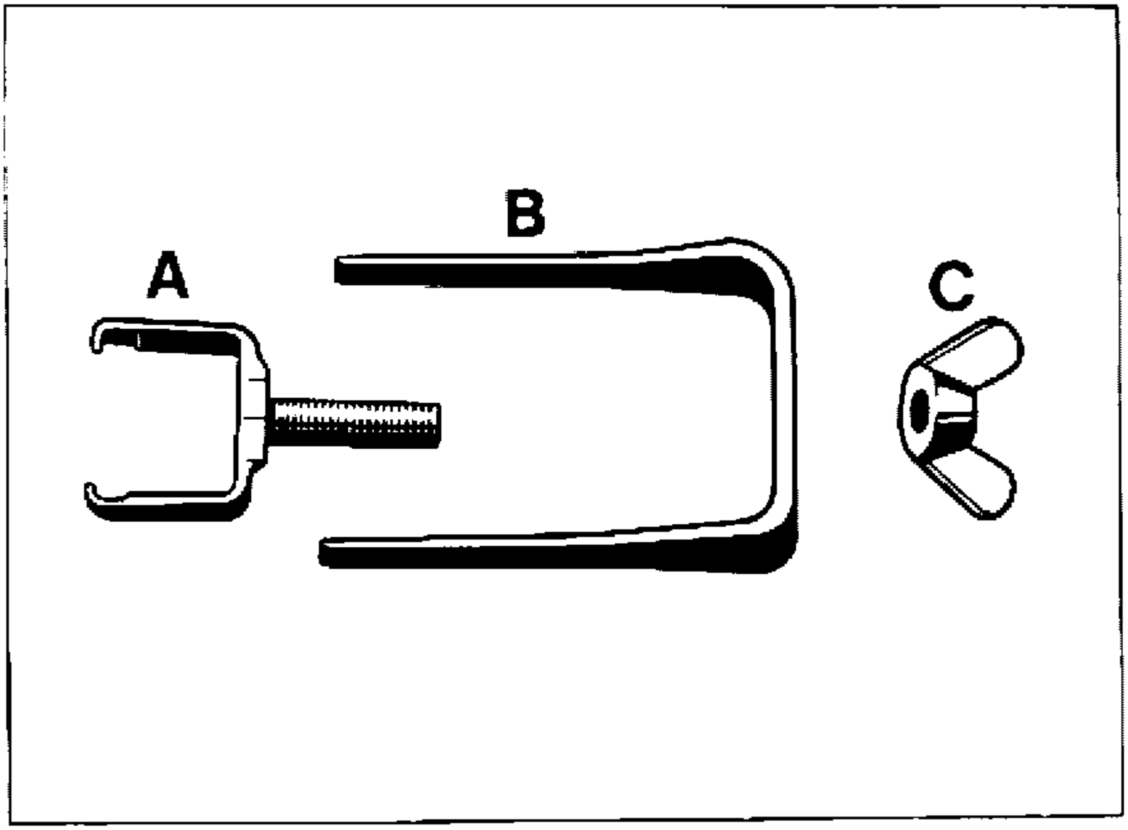


256-87

2. If the temperature mixing valve is not closed, undo the motor mount. Slide the motor in the slots until the temperature mixing valve is set in the closed end position.

85 30 19 Removing and installing heater fresh air blower motors

Tools



No.	Designation	Special tool	Order number	Explanation
	Puller assembly	9512	000.721.951.20	3 parts

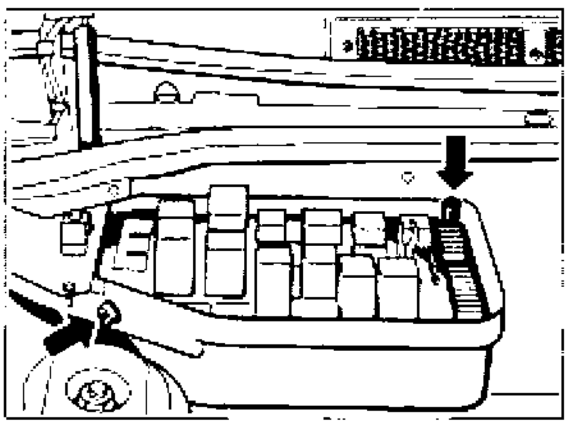
85 30 19 Removing and installing heater fresh air blower motors

Removing and installing heater fresh air blower motors with the heater/ A/C unit installed

1. Take off battery ground cable.

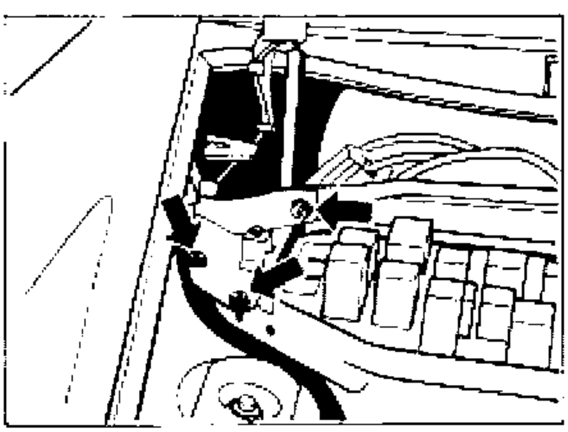
2. Remove cover of heater/ A/C unit (2 screws with washers).

3. Unbolt Central Electrical System.



240-85

4. Remove wiring harness cover.



241-85

5. Disconnect electrical connector from blower housing cover and place Central Electrical System on fender (use fender protectors).

6. Pull connector off the blower final stage.

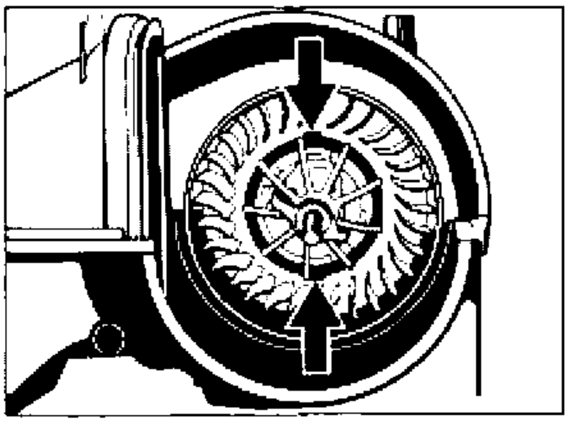
7. Remove firewall.

Right-hand blower

8. Separate connector and wire from housing cover. Unscrew knurled nut and use a screwdriver to detach retaining clamp from housing cover. Take off housing cover from above.

9. Take particle filter out of housing duct from above. Detach both mounting screws (Torx T 20) of blower motor.

10. Place part A of Special Tool 9512 onto shaft and turn right.
    Make sure the cutouts in the blower wheel line up with the cutouts in the blower housing.

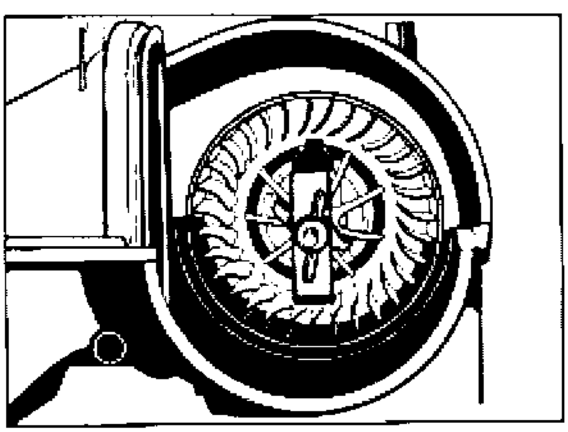


265-85

Note

Note the installation position for part A of special tool 9512. The hole for the fan wheel shaft is out of center (2 mm).

11. Push part B into the cutouts.



266-85

12. Tighten wing screw (part C) until the blower motor disengages.

13. Pull off connecting wire.

Left-hand blower

14. Detach reservoir from tank.

15. Pull wire off the housing cover. Screw out knurled nut and use a screwdriver to detach retaining clamp from housing cover. Take off housing cover from above.

16. For further steps, proceed as per items 9 to 13 above.

Note

When installing the motor, make sure the blower motor engages correctly and the connection wires are not squeezed.

Check if the blower wheel rotates freely.

Printed in Germany - 12, 1995

87 Pressure and temperature specifications

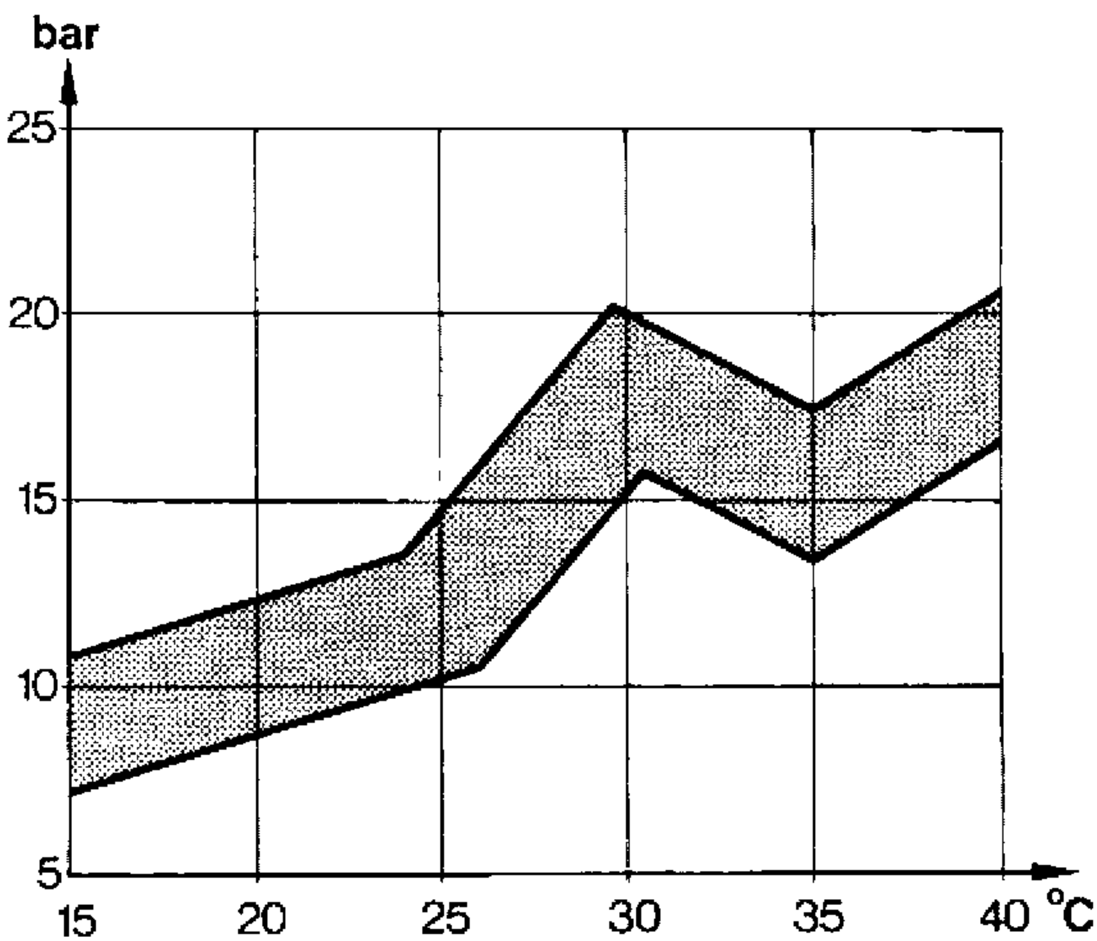
Refrigerant R 134 a

General test requirements:

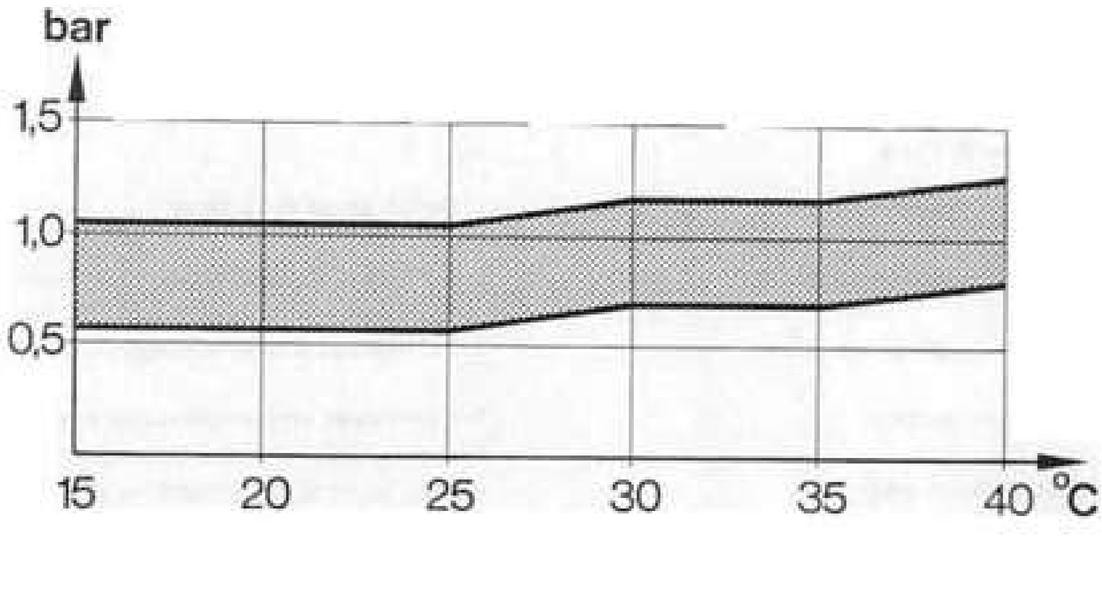
- V-belt tensioned correctly.
- Solenoid clutch is energized.
- Condenser is clean.
- Sunroof, doors and windows closed.

1. Switch on air conditioner.
2. Set temperature control to max. cooling.
3. Set fresh-air blower to stage 4.

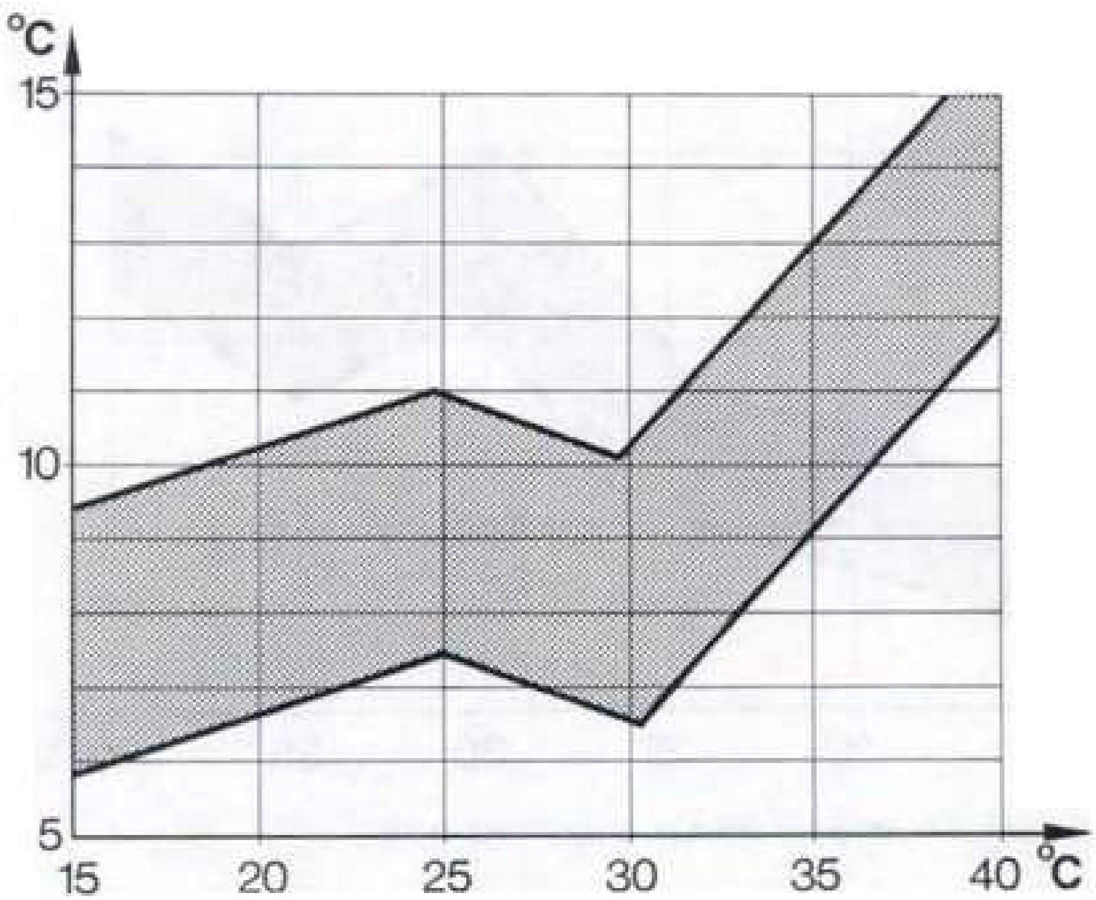
The pressures and temperatures specified in the below diagrams must be reached after approx. 10 mins. operation time at a speed of 2,000 rpm with the compressor switched on.



High pressure in refrigerant circuit vs. ambient temperature

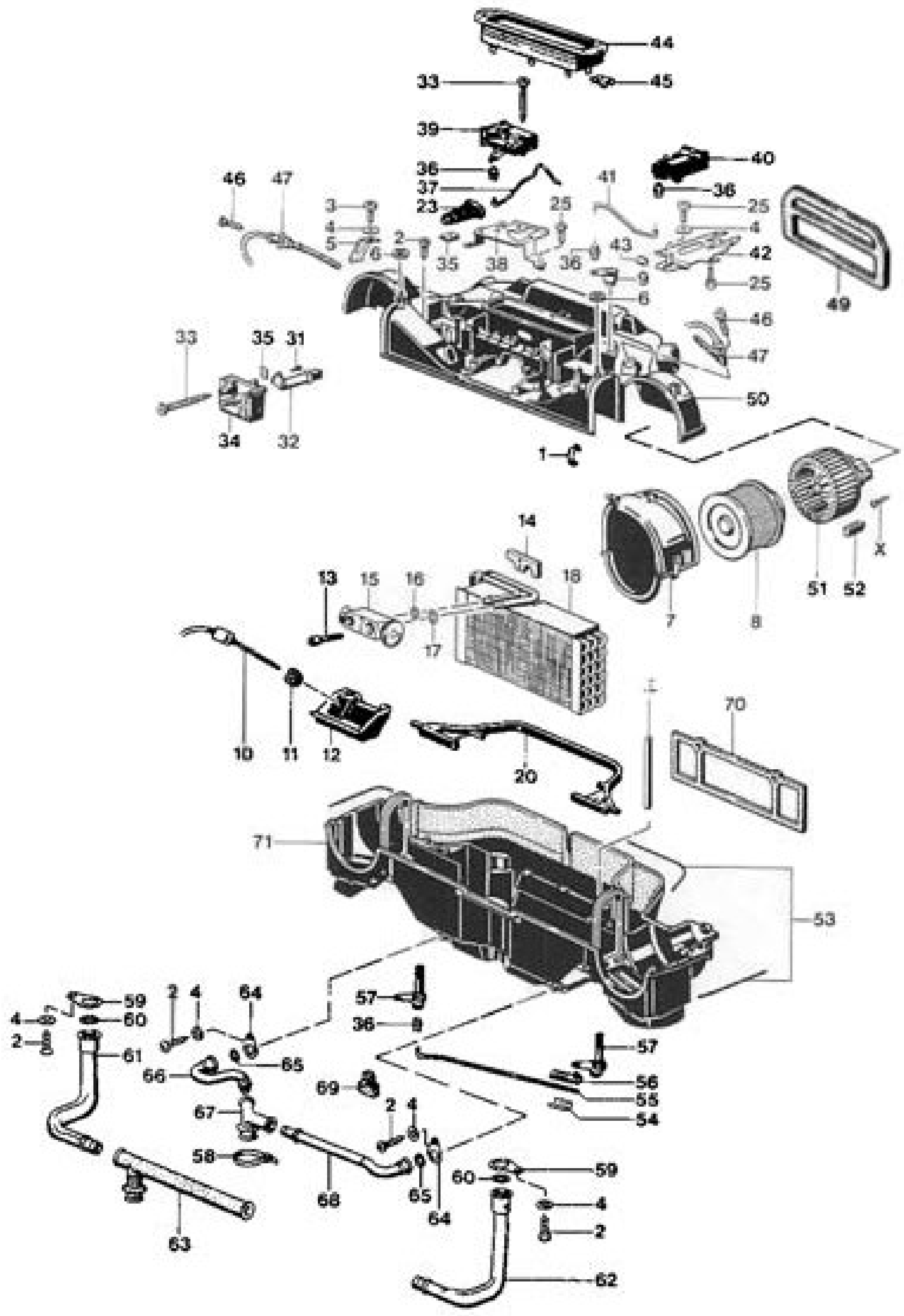


Low pressure in refrigerant circuit vs. ambient temperature



Air temperature at center nozzle vs. ambient temperature

87 01 19 Dismantling and assembling heater/ A/C unit



No.	Designation	Qty.	Note: Removal	Installation
1	Tensioning spring	32		
2	Screw	11		
3	Screw	9		
4	Washer	6		
5	Bracket	1		
6	Knurled nut	2		
7	Housing cover LH	1		
	Housing cover RH	1		
8	Particle filter LH	1		
	Particle filter RH	1		
9	Lever	1		
10	Evaporator temperature sensor	1	Built into wiring harness	Built into wiring harness
11	Grommet	1		
12	Cover			
13	Hexagon socket head screw M 5	2		
14	Retaining plate	1		
15	Expansion valve	1		
16	Gasket	1		Replace. Apply a thin coat of refrigerant oil.
17	Gasket	1		Replace. Apply a thin coat of refrigerant oil.
18	Evaporator	1	Do not damage fins	Do not damage fins
19	Link	1		
20	Gasket	1		
21	Deleted			
22	Deleted			

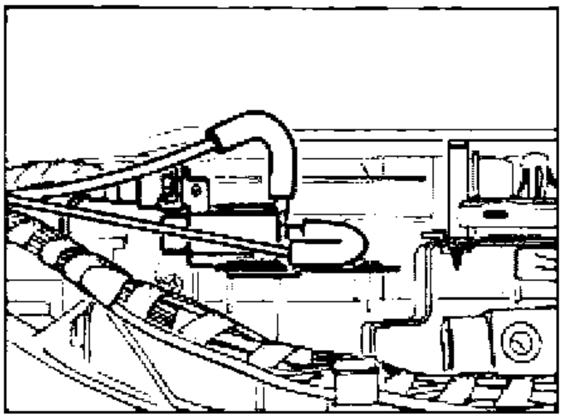
No.	Designation	Qty.	Note: Removal	Installation
23	Bellows	1		
24	Deleted			
25	Screw	5		
26	Deleted			
27	Deleted (Temperature mixing flaps are located below the instrument panel in the driver and passenger footwells)			
28	Deleted			
29	Deleted			
30	Deleted			
31	Screw	1		
32	Joint	1		
33	Screw	4		
34	Motor for defroster or center nozzle	1		For adjustment, refer to page 85-1
35	Sheetmetal nut	4		
36	Linkage clip	5		
37	Linkage	1		
38	Bracket	1		
39	Motor for fresh-air flap	1		
40	Motor for footwell flaps	1		For adjustment, refer to page 85-1
41	Linkage	1		
42	Bracket	1		
43	Retaining clamp	3		
44	Sheetmetal frame with rubber bellows	1		
45	Sheetmetal nut	2		

No.	Designation	Qty.	Note: Removal	Installation
46	Screw	2		
47	Temperature sensor, mixing chamber	2	Built into wiring harness	Built into wiring harness
48	Deleted			
49	Gasket	1		
50	Housing upper section	1		
51	Blower LH			White plastic
	Blower RH	1		Black pastic
X	Screw (Torx T 20)	4		
52	Rubber mount	8		
53	Sealing ring	1		
54	Retaining clamp	1		
55	Linkage	1		
56	Lever	1		
57	Drive	2		
58	Tie-wrap	2		
59	Bracket	2		
60	Sealing ring	2		
61	Water drain tube	1		
62	Water drain tube	1		
63	Spacer	1		
64	Bracket	2		
65	Seal	2		
66	Water drain tube	1		
67	Bracket	1		
68	Water drain tube	1		
69	Rubber mount	1		
70	Gasket	1		
71	Housing lower section	1		

Modifications to heater / A/C unit from model year '95

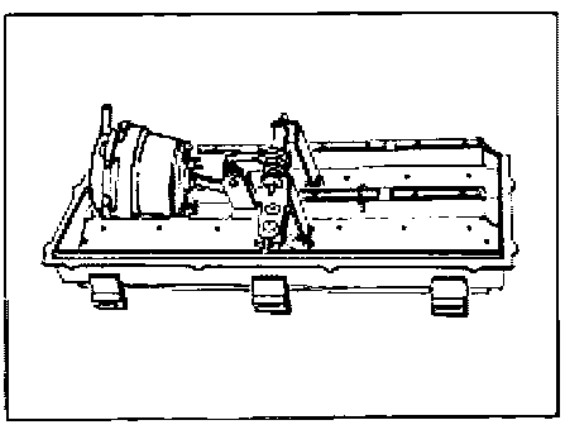
Electrically controlled bypass air flaps

A solenoid valve controls the vacuum capsule of the bypass air flaps. The solenoid valve is mounted on the right of the heater / A/C unit.



2181-87

As a spare part, the vacuum capsule with bypass air flaps is only available as a complete unit. If you need to remove the vacuum capsule with bypass air flaps, the heater / A/C unit must first be emptied and removed.



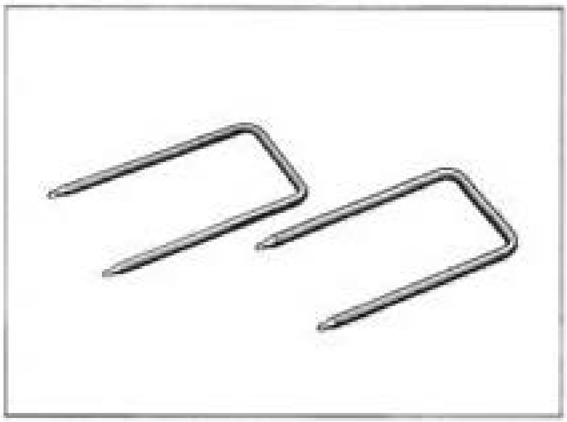
2182-87

87 02 19 Removing and installing heating and air conditioning control

Note

Use Special Tool V 160 (this code also is the Part No.) to remove the heating and air conditioning control.

Supplier: Matra Werke GmbH
Dieselstrasse 30 - 40
D-60314 Frankfurt



254-87

1. Insert and engage Special Tool in cutout provided in the heating and air conditioning control.

2. Push out heating and air conditioning control manually from below towards the rear.

3. Disengage 25-pin and 35-pin connector.

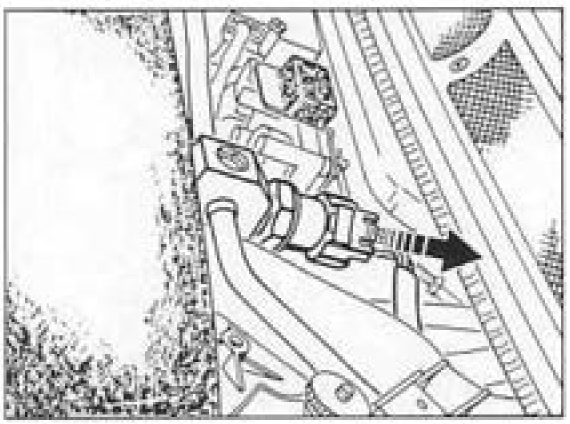
87 83 19 Removing and installing A/C system pressure switch

Removal

1. Draw off refrigerant charge with service equipment.

2. Remove cover of heater/ A/C unit (2 screws with washers).

3. Pull off connector at pressure switch.



1796-87

4. Loosen and screw out pressure switch.

Installation

5. Replace O-ring at pressure switch and coat lightly with refrigerant oil.
   Tightening torque: 3 Nm (2 ftlb.)

87 03 Safety instructions for handling refrigerant R 134 a

The refrigerant R 134a used in this system is specified as a safety refrigerant. This means that this refrigerant is inflammable, non-explosive, non-toxic, non-irritating, odorless and tasteless. The following precautions should nevertheless be observed.

1. Avoid any contact with liquid or gaseous refrigerant. Treat all skin areas affected as in the case of frostbite; rinse with cold water immediately and consult a doctor. Wear goggles to protect your eyes. Consult a doctor immediately if refrigerant gets into your eyes. Wear protective gloves to protect your hands.

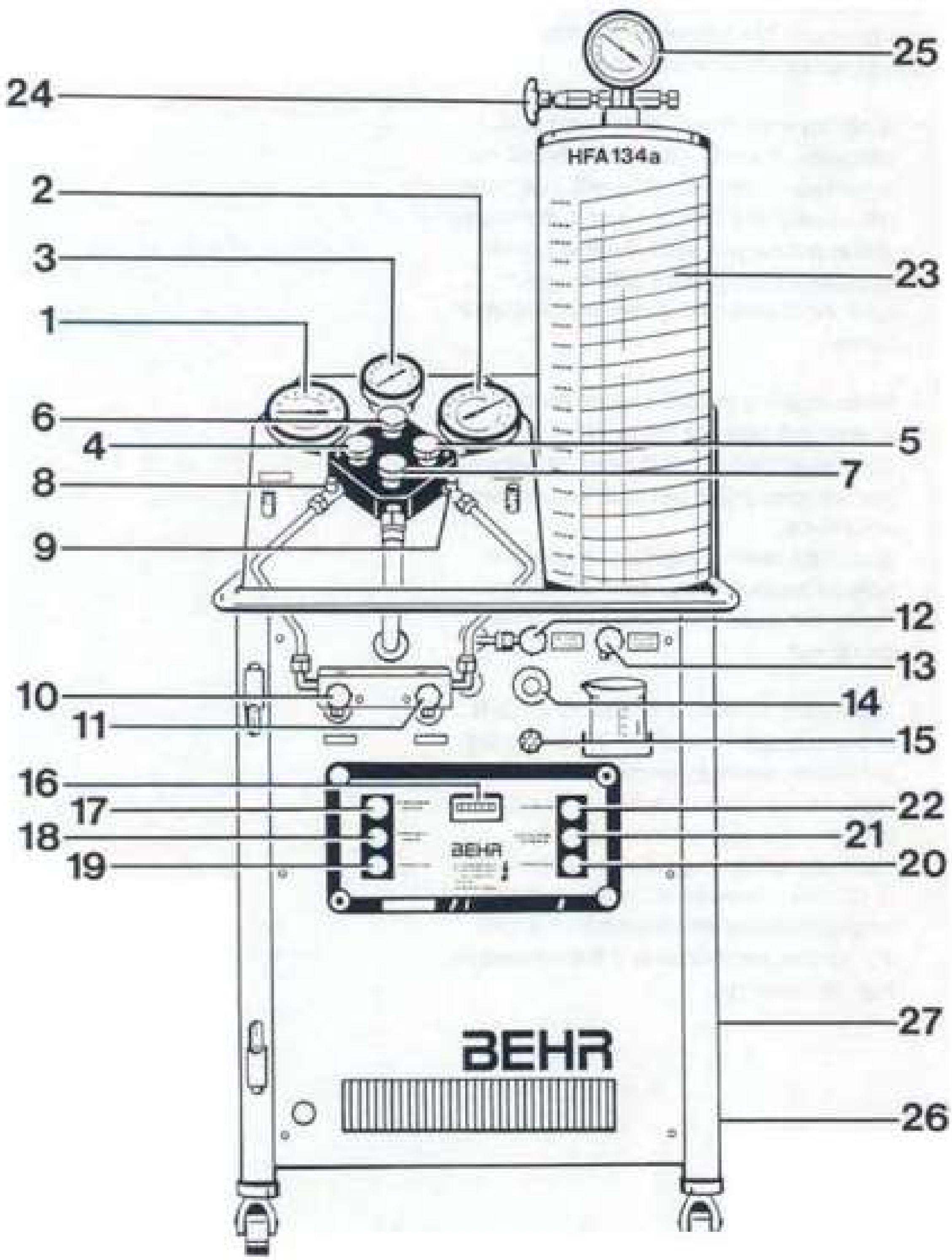
2. When repairing the A/C system, drain the system and clean the refrigerant. Even non-chlorine refrigerants must never be released into the atmosphere and must be disposed of correctly.
   Due to the differing chemical composition, different types of refrigerant must never be mixed with each other (not even in small quantities).

3. Never perform welding operations on parts of the closed A/C system or in their immediate vicinity. Heating generates excessive pressure regardless of whether the system is filled with refrigerant or not and may cause system damage or even explosions. R 134 a is completely non-toxic at normal temperatures but will decompose if it gets into contact with flames or if it is exposed to high temperatures.

4. Do not throw refrigerant bottles and do not expose filled bottles to direct sunlight or other heat sources for longer periods of time. The maximum admissible temperature of a filled refrigerant bottle must not exceed 45 °C.

87 03 17 Servicing the A/C system

Service equipment SECU 134



1 - Low-pressure gauge

2 - High-pressure gauge

3 - Torr gauge

4 - Low-pressure shutoff valve (blue)

5 - High-pressure shutoff valve (red)

6 - Torr gauge shutoff valve (black)

7 - Vacuum pump shutoff valve (yellow)

8 - Low-pressure fitting

9 - High-pressure fitting

10 - Shutoff valve for refrigerant inlet

11 - Shutoff valve for refrigerant outlet

12 - Shutoff valve for refrigerant oil inlet

13 - Refrigerant oil drain valve

14 - Humidity indicator

15 - Vacuum oil tank fitting

16 - Operating hours gauge

17 - SUCTION FINISH (ABSAUGEN ENDE) warning lamp

18 - SUCTION/PURGE (ABSAUGEN/ REINIGEN) pressure switch

19 - ON/OFF main switch

20 - HEATING (HEIZUNG) pressure switch

21 FAULT (STÖRUNG) warning lamp

22 - VACUUM PUMP (VAKUUMPUMPE) pressure switch

23 - Filling cylinder with weight scales

24 - Filling cylinder shutoff valve

25 - Filling cylinder high-pressure gauge

26 - Refrigerant oil inlet flange

27 - Refrigerant oil accumulator inspection glass

Note

Observe manufacturer's operating instructions or repair instructions when working on the service equipment.

Assembly operations that require opening the refrigerant system

The system contents must be disposed of properly after all operations on the A/C system that require opening the refrigerant system. Make sure the relevant safety regulations are observed.

To keep dirt and humidity out of the A/C line system, be careful to ensure absolute cleanliness whenever working on the system. Never clean any parts of the system with hot steam. Use only nitrogen for cleaning.

When replacing a component, plug all openings with suitable plugs.

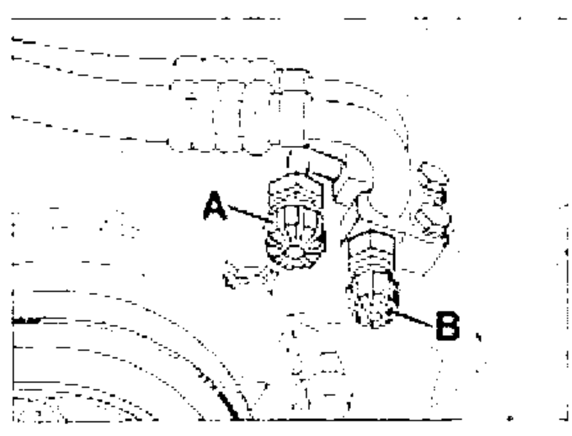
Never use refrigerant R 12 to fill or top up an A/C system designed for R 134 a, not even vice-versa (this is due to the different components and refrigerant oils used).

General sequence of assembly:

1. Draw off refrigerant.
2. Remove failed component.
3. Evacuate.
4. Check system for tightness.
5. Flush with refrigerant.
6. Draw off system contents once more.
7. Evacuate.
8. Fill.

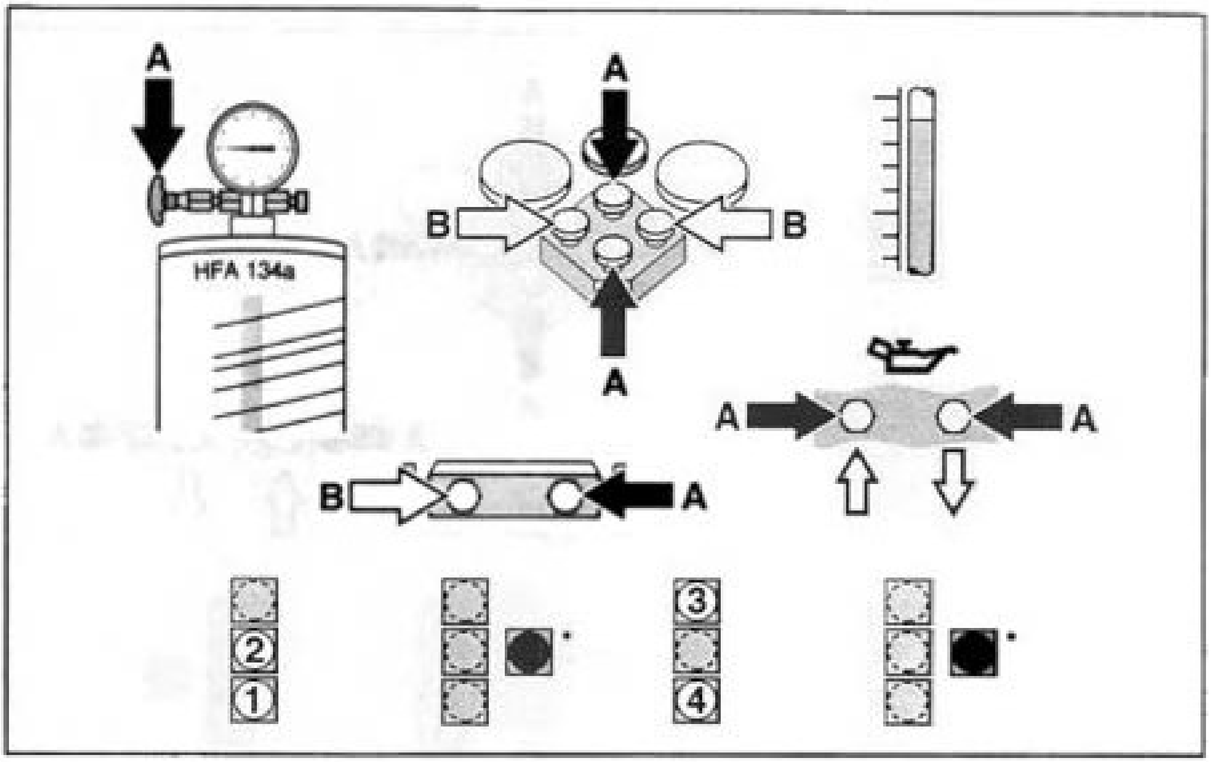
Note

Check seals when disconnecting or reconnecting hose connections.



A - High pressure
B - Low pressure

Drawing off refrigerant and cleaning system



Note

Close all valves before each step.

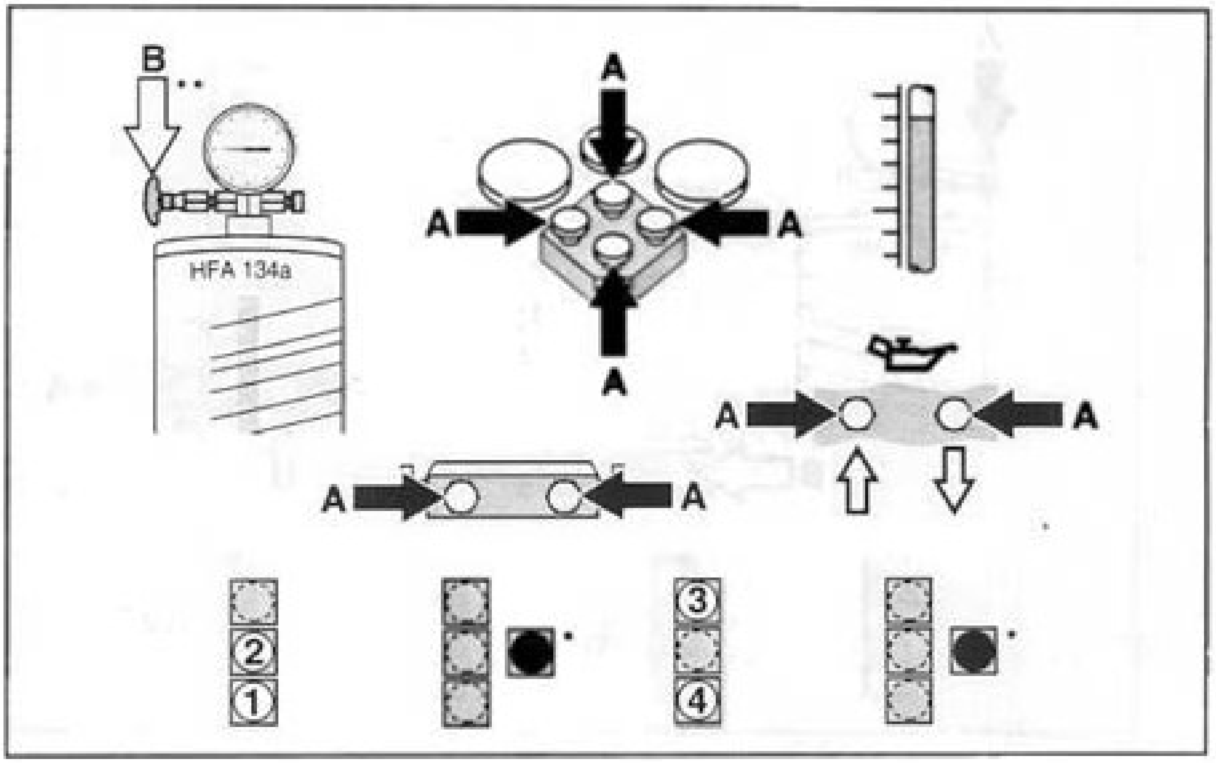
A - Close B - Open

Drawing off: Start End

1 - ON / OFF 3 - SUCTION / END
2 - SUCTION / PURGE 4 - ON / OFF

* Pressure too high / filling cylinder full.
Drain refrigerant from filling cylinder into refrigerant bottle (approx. 50 %).

Cleaning refrigerant



A - Close B - Open

Cleaning: Start

1 - ON / OFF
2 - SUCTION / PURGE

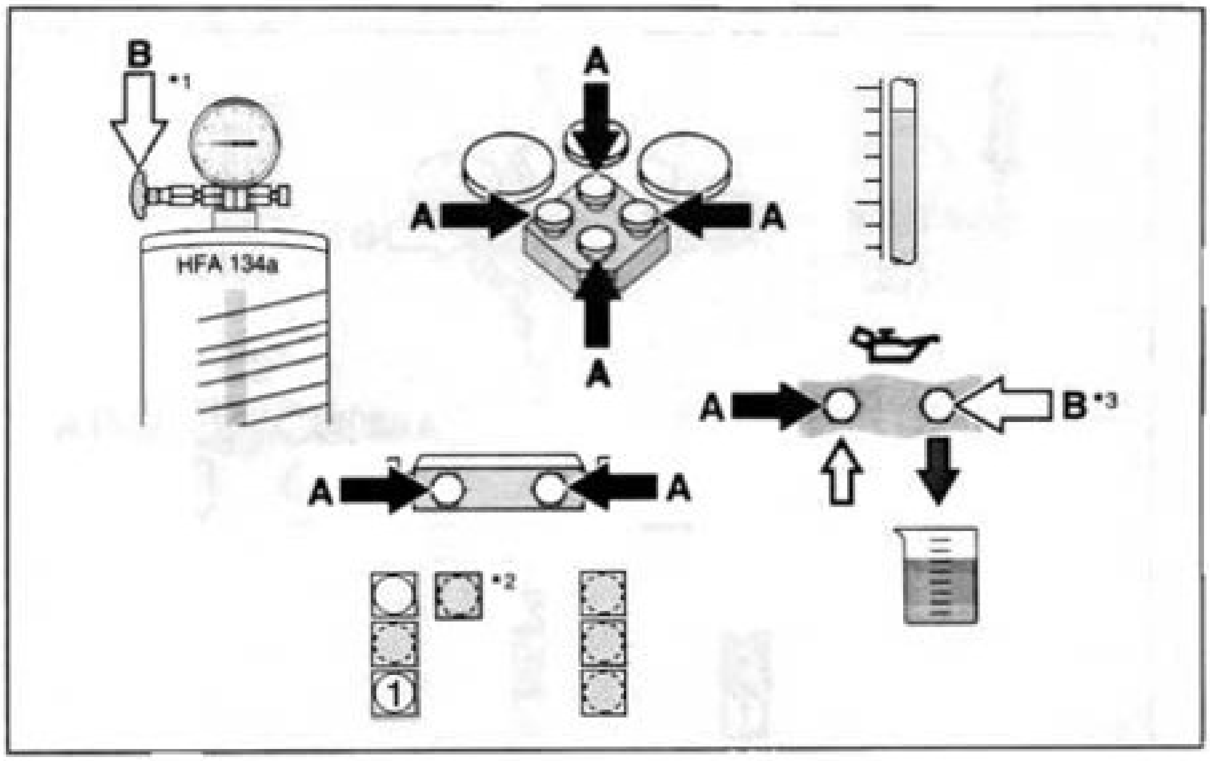
End

3 - PURGE / END
4 - ON / OFF

Pressure too high / filling cylinder full.

OPEN one turn. CLOSE after approx. 15 minutes.

Draining old refrigerant oil



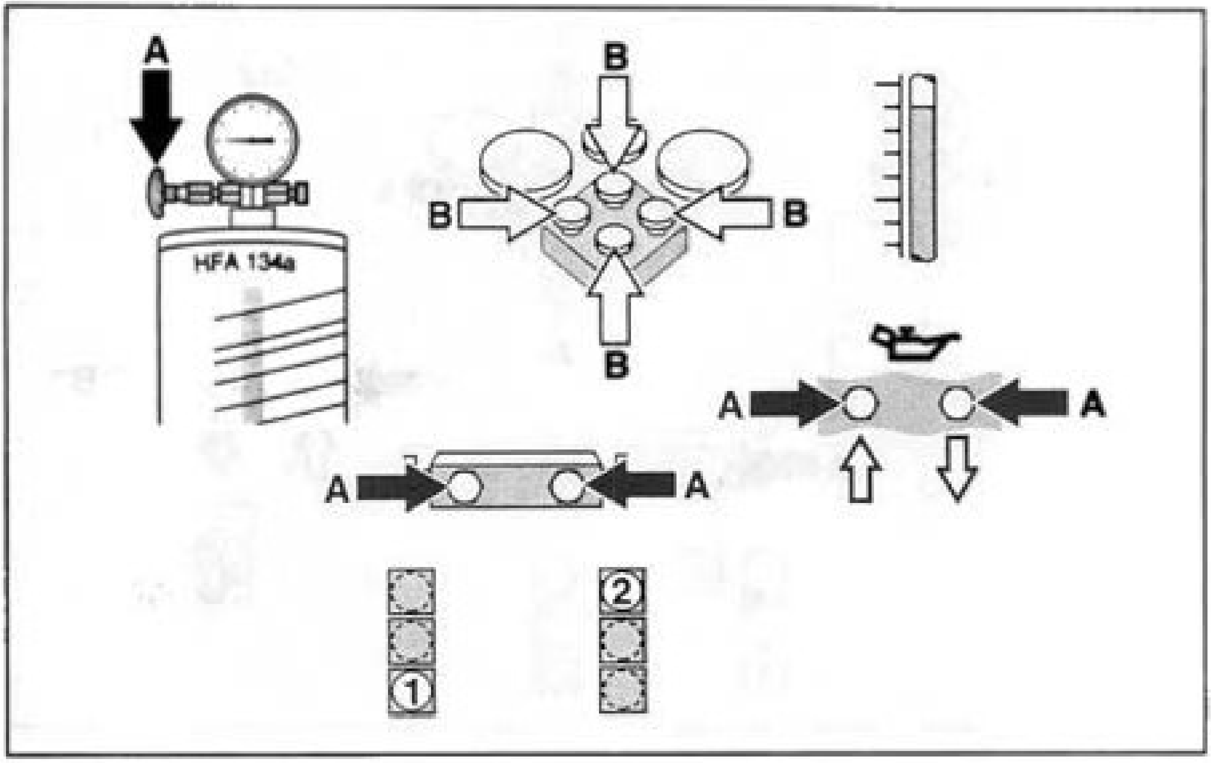
A - Close B - Open

1 - ON / OFF

Instructions

* 1 open until *2 OFF, then *1 CLOSED and 3* OPEN.

Evacuating

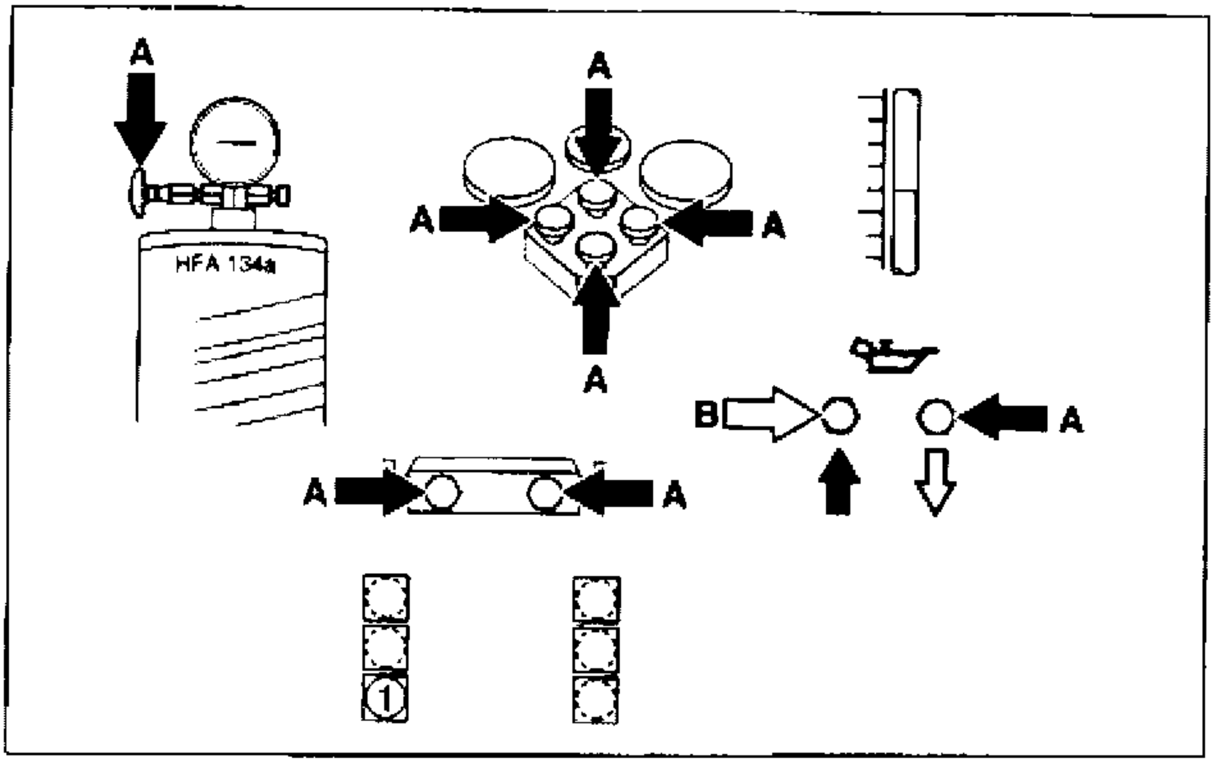


A - Close B - Open

1 - ON / OFF
2 - VACUUM PUMP

Minimum evacuating time: 15 minutes.

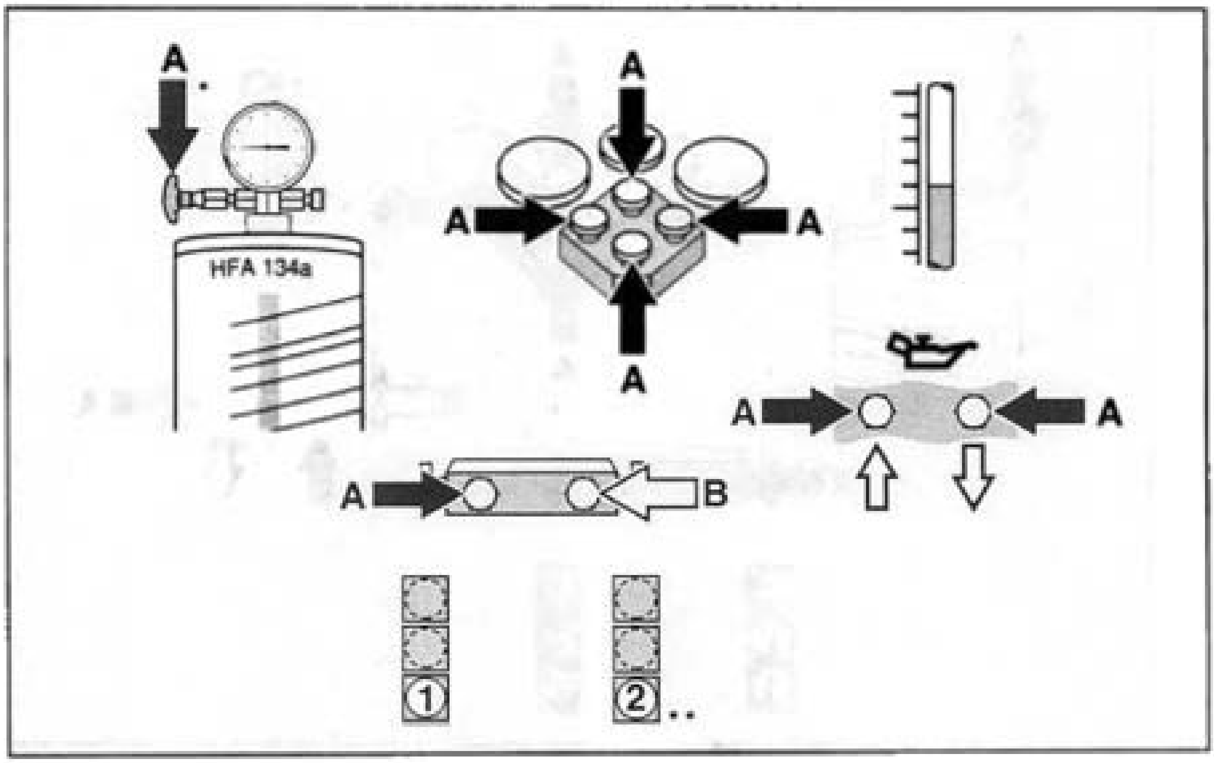
Topping up with new refrigerant oil



A - Close B - Open

1 - ON / OFF

Filling from high-pressure side



A - Close B - Open

1 - ON/OFF
2 - HEATING

* Pressure: 8 to 10 bar.

** If pressure is below 8 bar.

Draw off internally after using the SECU (page 87-17).
Close manual valves on both hose lines first.

Refilling A/C unit

Note

The A/C unit is no longer fitted with an inspection glass for refrigerant level. If the unit develops insufficient cooling power, the refrigerant must be removed, the specified refrigerant volume filled in and the unit tested for tightness.

1. Remove refrigerant using service unit.

2. Determine refrigerant oil concentration in removed refrigerant.

3. Fill in new refrigerant oil.

4. Evacuate.

5. Fill in specified volume of refrigerant.

6. Check system for tightness.

Distribution of oil quantity in refrigeration circuit

Total oil quantity	**140 ± 20 cc**

After the system contents have been drawn off, the following oil quanitites remain in the system:

Condenser	approx. **20 cc**
Evaporator	approx. **30 cc**
Fluid reservoir including piping	approx. **10 cc**
Compressor	approx. **50 cc**
Oil quantity circulating in refrigeration circuit	approx. **40 cc**

Note

On **new vehicles,** the drawn-off oil quantity is approx. **15...20 cc**.
On **older vehicles,** this oil quantity is higher. The oil quantity drawn off must be added to the system again.

Refrigerant oil drawn off from an A/C unit that has already been operated must **not** be reused. When replacing a component, the oil quantity must be topped up by the amount remaining in the removed component.

When the A/C system is **depressurized,** the refrigerant oil circulating in the circuit (approx. **40 cc**) will be lost and must be added to the A/C system before filling up with refrigerant.

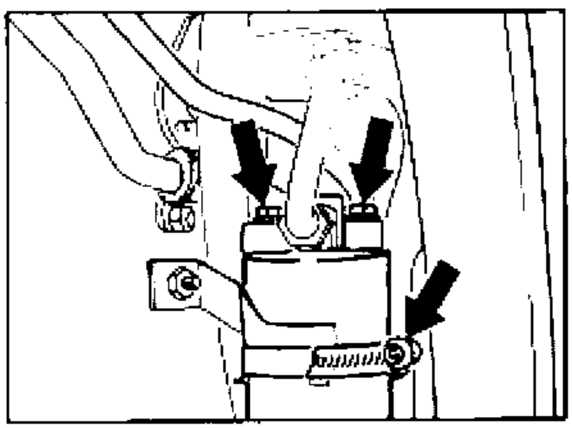
87 55 19 Removing and installing liquid tank

Removal

1. Remove refrigerant using service unit.

2. Remove rear wheel housing liner from front left fender.

3. Loosen hose clamp on liquid tank. Unscrew the two hexagon head screws on the liquid tank and lower the liquid tank out. Insert air-tight plugs in liquid lines immediately.



1897-87

Installation

1. Do not remove plugs until just before installation. Replace O-ring on branch piece and refrigerant line and wet O-ring with refrigerant oil.

2. Tighten the two hexagon head screws on the liquid tank to 10 Nm (7 ftlb.).

3. Determine refrigerant volume and top up with refrigerant oil **(volume drawn off + 10 cc).**

Note

The reservoir must be replaced if system malfunctions occur (e.g. due to accident damage or loss of air conditioning system pressure). The refrigerant sight glass has been deleted.

Refrigerant oil drawn off from an A/C system that has already been in operation must not be reused.

87 50 19 Removing and installing condenser

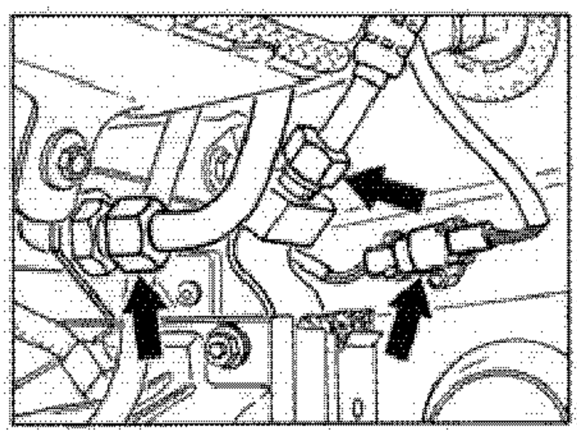
Removal

1. Use service unit to draw off refrigerant.

2. Unbolt and take off left front wheel.

3. Remove wheel housing liner and lower section of front spoiler.

4. Disconnect connector linking the ballast resistor to the condenser fan. Undo refrigerant pipes from condenser, using a suitable tool to lock.



1898-87

5. Plug ports and pipes immediately with a suitable air-tight seal.

6. Pull off electrical connector from fan motor. Unclip headlight wire and vent hose.

7. Disconnect both headlight cleaner hoses from regulator valve. Detach condenser support bracket from lower body end and remove as a unit.

8. Undo self-tapping screws (4 ea.) from support bracket and pull out condenser from above.

Installation

1. Remove plugs inserted in pipes and condenser port only imediately before refitting components.

2. Replace O-rings and coat lightly with refrigerant oil.

3. **Tightening torques** for
   refrigerant pipes:
   5/8" - 18 UNF **17 ± 3 Nm (12.5 ± 2)**
   3/4" - 16 UNF **24 ± 4 Nm (18 ± 3)**

4. Determine refrigerant oil volume and refill with refrigerant oil (**drawn off volume + 20 cc**).

Note

Refrigerant oil drawn off from an A/C system that has already been operated must not be reused.

87 53 19 Removing and installing condenser fan

Removal

1. Remove lower front spoiler section on the left side.

2. Pull off electrical connector from fan motor. Unclip headlight wire and vent hose.

3. Pull both headlight washer hoses off the regulator valve. Remove fan bracket from condenser retaining bracket and take out from below.

4. Detach fan from retaining bracket and take off fan.

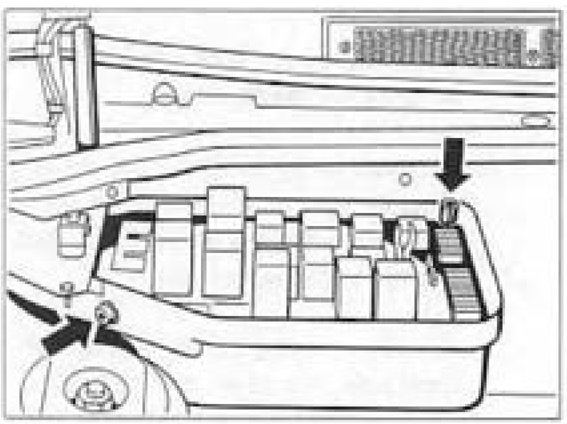
Note

Check after installation if fan is able to rotate freely.

87 01 19 Removing and installing heater / air conditioning unit

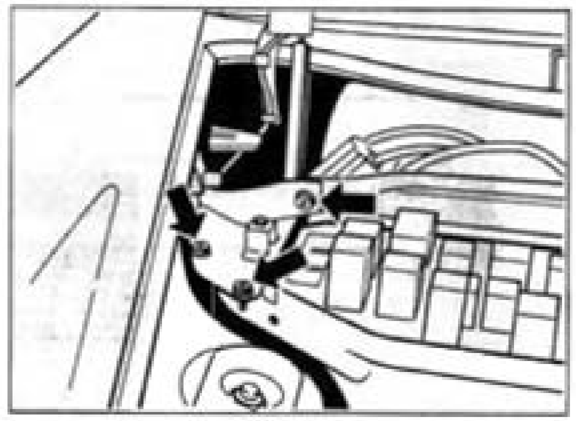
Removal

1. Take off battery ground cable.
2. Draw off refrigerant with service equipment.
3. Remove fuel tank.
4. Remove cover of heater/ A/C unit (old version is held in place by 2 bolts with washers, new version is fitted with spring clamp).
5. Remove Central Electrical System.



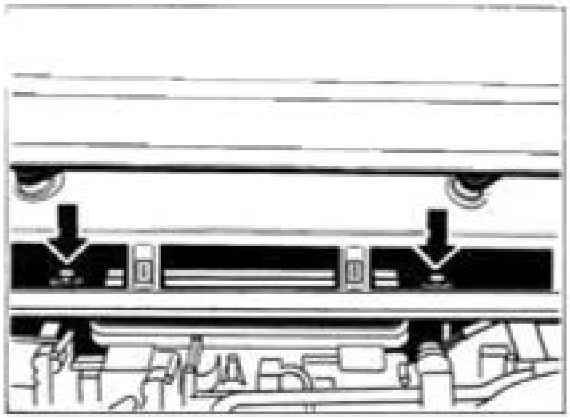
240-87

6. Remove wiring harness cover.

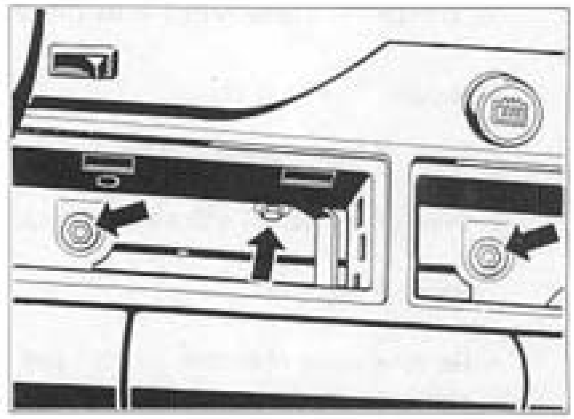


241-87

7. Detach electrical connector from fan housing cover and place Central Electrical System on fender (use fender protectors).
8. Detach wire connector from blower final stage.
9. Remove firewall.
10. Remove fresh-air cleaner grille.
11. Undo fastening screws.



242-87



250-87

12. Disconnect air conditioning pipes from expansion valve. Plug ports and piping immediately with air-tight seals.

13. Pull off connector from pressure switch.

14. Detach A/C piping bracket and move A/C piping out of the way.

15. Pull off condensation water and water drain hoses (disconnect tie-wraps) and push rubber grommets forward.

16. Remove radio, heater/ A/C regulator and glove compartment. Pull off connector above glove compartment.

17. Undo fastening bolt and nuts.

18. Remove side panel and pull off right and left hot air induction flanges.

19. Pull off connector from servo motor for temperature mixing valves.

20. Separate wiring harness from heater/ A/C regulator (harness is tied to the passenger compartment wiring harness) and pull through to passenger side.

21. Start by removing heater/ A/C unit on right-hand side. Pull off air ducting to side nozzles and pull wiring harness along.

Installation

1. Detach spring clamp of left and right heater/ A/C unit (new version).

2. When re-installing, fit right-hand air guide before heater/ A/C unit is brought into installation position (for better accessibility). The left-hand air guide can be pushed in place with the heater/ A/C unit fitted after the large instrument cluster has been removed.

3. Replace O-ring for refrigerant pipes at expansion valve and coat lightly with refrigerant oil.

Note

Be sure to top up with refrigerant oil when replacing the heater/ A/C unit or the evaporator. Determine refrigerant oil volume and refill with refrigerant oil (**volume drawn off + 30 cc**). Refrigerant oil drawn off from an A/C system that has already been operated must not be reused.

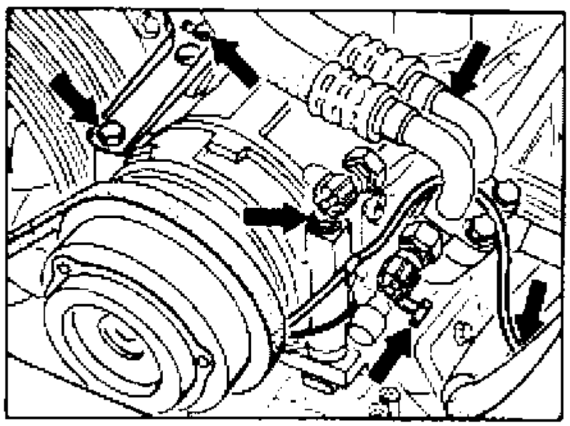
87 34 19 Removing and installing compressor

Removal

1. Draw off refrigerant with service equipment.

2. Pull off wiring connector.

3. **Only slacken** compressor mounting bolts. Undo tensioning bolt lock nut and screw out tensioning bolt.



1477-87

4. Detach refrigerant pipes from compressor. Plug ports and piping immediately with air-tight seals.

5. Move compressor to the left and take off V-belt. Take out mounting bolts and take off compressor.

Installation

Note

New compressors are pressurized and are filled with the respective total oil quantity required for the refrigerant circuit. The oil quantity remaining in the individual components must therefore be taken into account.

1. Screw cap off the service valve and release pressure from compressor. Detach service valve from compressor (4 bolts) and take off valve.

2. Pour approx. **50 cc** of refrigerant oil from the compressor into a graduated measuring glass. The remaining oil quantity (approx. **90 cc**) remains in the compressor.

Note

Refrigerant oil drawn off from an A/C system that has already been operated must not be reused.

3. **Tightening torque** of compressor mounting bolts (M 8 x 125 bolts): **28 Nm (20.5)**

4. **Tightening torque** of refrigerant pipe (M 8 x 32 bolts): **23 Nm (17)**
   Replace all mounting bolts during each reassembly (micro-sealed bolts).
   Replace O-rings and coat lightly with refrigerant oil.

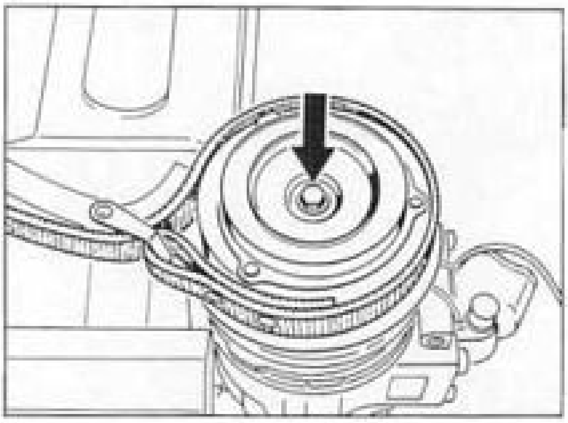
5. **Tightening torque** of service valve (4 bolts): **25 Nm**

6. Remove plugs from piping and compressor port only immediately before refitting.

87 27 19 Removing and installing magnetic clutch

Removal

1. Use a standard strap wrench to keep thrust plate from turning and undo mounting bolt.



1927-87

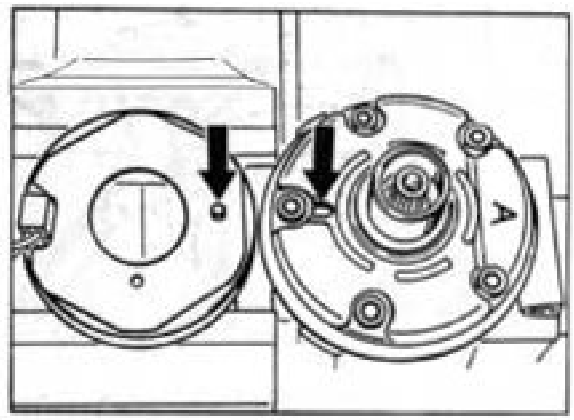
2. Screw a M 8 bolt into the thrust plate threads until the thrust plate can be taken off by hand.

3. Take out spacers and remove circlip with a pair of standard circlip pliers. Pull off pulley manually.

4. Unscrew solenoid coil wire from compressor housing. Remove circlip. Take solenoid coil off the compressor housing.

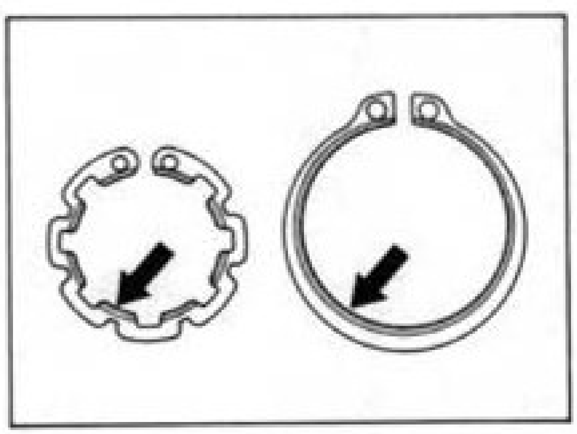
Installation

1. Place solenoid coil into compressor housing. Lock pin must engage into the lock pin hole.



1928-87

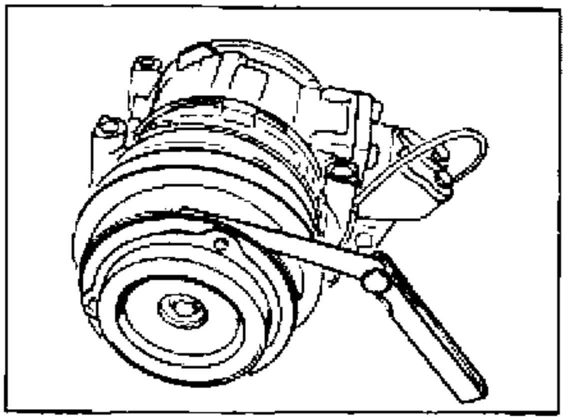
2. Fit circlips. The chamfered face of the circlip points upward (towards mounting bolt).



1930-87

3. Tighten thrust plate mounting bolt. **Tightening torque: 14 Nm (10)**

4. Use a feeler gauge set to check air gap between thrust plate and pulley.
   Air gap: 0.5 mm ± 0.15 mm



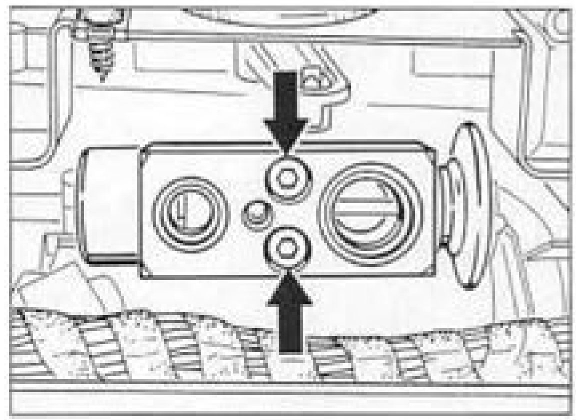
1929-87

If the air gap is outside the limits, adjust air gap by inserting or removing spacers.

87 70 19 Removing and installing expansion valve

Removal

1. Take off battery ground cable.
2. Draw off refrigerant with service equipment.
3. Remove heater/ A/C unit cover (old version is held in place by 2 bolts with washers, new version is fitted with spring clamp).
4. Remove Central Electrical System.
5. Remove wiring harness cover.
6. Detach electrical connector at fan housing cover and place Central Electrical System on fender (use fender protectors).
7. Detach connector from blower final stage.
8. Remove firewall.
9. Separate A/C piping from expansion valve. Plug ports and piping immediately with air-tight seals.
10. Undo and take out expansion valve. Plug piping to evaporator immediately with air-tight seals.



1966-87

Installation

Remove plugs only directly before refitting. Replace O-rings and coat lightly with refrigerant oil.

Refrigerant oil drawn off from an A/C system that has already been operated must not be reused.

Tightening torques:

M 5 bolt **6 Nm (4.5)**
M 6 bolt **9 Nm (6.6)**

90 73 Tuning model '94 remote control units

Vehicles with immobilizer M 530

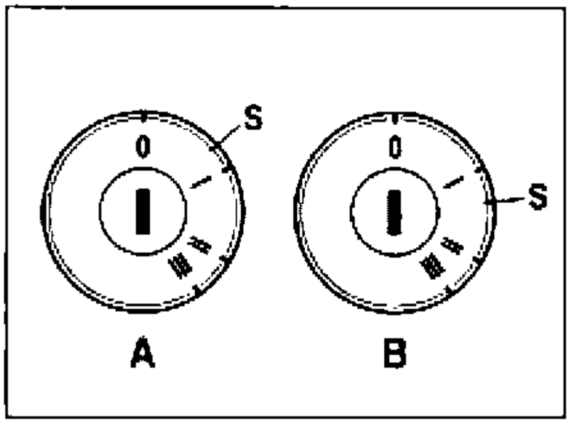
Note

As a result of tolerances in the ignition lock, it is not possible to define precisely the position at which the buzzer contact is closed by the ignition key.
Depending on the tolerances, the buzzer contact(S) in the ignition lock

may be closed between positions 0 and I (Fig. A)

or

between positions I and II (Fig. B).



2213-90

I = terminal X (windshield wipers on)
II = terminal 15 (ignition on)
III = terminal 50 (starter)

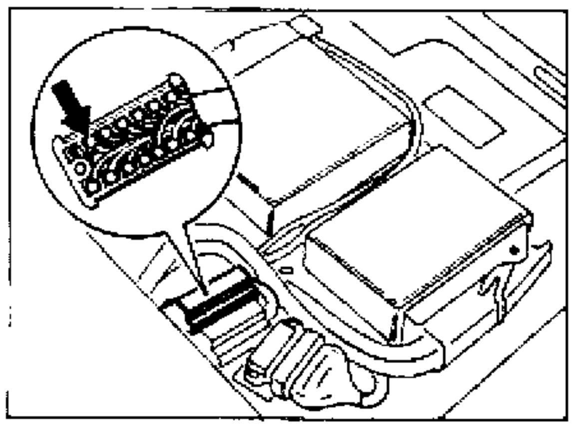
When the key is removed from the ignition lock, the buzzer contact is opened (switched off). This signal is required for tuning the remote control units.

The position where the buzzer contact is operated can be located as follows.

Manual transmission vehicles.

The buzzer contact signal is transmitted via pin 12 in the 14-pin connector (passenger compartment wire harness, connector X4/1 under the left seat).

1. Remove the left seat. Do not unplug the connector; carefully remove the connector cover and connect a test lamp to pin 12 (red/black) and ground.



2214-90

2. Turn the key slowly from position 0 to position II in the ignition lock and observe the position at which the test lamp lights up.

Tiptronic vehicles:

The buzzer contact signal can be observed on the
LEDs of the selector indicator (speedometer instrument cluster).

1. Turn the key slowly from position 0 to position II in the ignition lock.

2. Observe the position at which one of the LEDs of the selector indicator lights up.

Important!

Manual and Tiptronic vehicles:

Depending on the position of the buzzer contact, for tuning (point 6 on page 3), the ignition key must be turned to

– position I (windshield wipers on)

or

the buzzer contact position (S) (see Fig. 2213-90 A and B).

Fig. A: buzzer contact (S) between position 0 and I. For tuning, the key must be turned to position I (windshield wipers on).

Fig. B: buzzer contact(S) between positions I and II. In this case, the key must be turned to the buzzer contact position (test lamp or selector indicator) for tuning.

Exception!

If the buzzer contact is reached just before or directly at "ignition on":
...tuning is not possible. In this case contact Porsche, Dept. VKG.

Tuning:

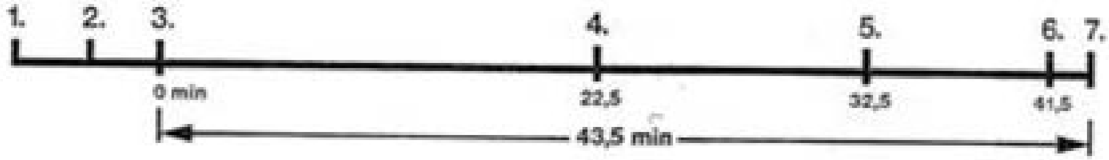
Tiptronic vehicles:

Before tuning the remote control units, check the key position at which the buzzer contact is operated. Then tune the units (see pages 3 and 4).

Manual vehicles:

Try to tune the units first.
Reason: In the majority of cases, the buzzer contact will be operated between ignition key positions 0 and I.
If it is not possible to tune the units on a manual vehicle, a test lamp must be used to locate the buzzer contact position.
Tuning must then be repeated.

Tuning the remote control units



The diagram shows the sequence of events. The time required is at least 43,5 minutes.

1. **Unlock** the vehicle **using the remote control unit** and open the door. Do **not** insert the key into the ignition lock (the door of the vehicle may be open or closed.)

2. Wait **3 minutes.** After this time, the immobilizer will **prime itself.** In order to ensure that you comply with the times given below, please use a stopwatch or alarm clock. If necessary, write the times down.

   Note:

   When the instructions below require "switch ignition on and off", it is sufficient to turn the ignition key until the ignition is switched off.

3. Now **start the stopwatch and at the same time switch the ignition on.** Leave the ignition key turned to "**ignition on**". Switching the ignition on starts a sequence of operations. After 41.5 minutes (± 1 Min.), the system reaches the status "tuning of remote-control units."

4. After **22.5 minutes** (± 2 Min.), the **ignition** must be switched off and on again.

5. After a total time of **32.5 minutes** (± 2 Min.), the **ignition** must again be switched off and on. Then lift the **windshield wipers** off the windshield (to detect ignition key position I (see point 6).

6. After a total time of **41.5 minutes** (± 1 Min.), perform the following sequence, starting with the ignition on:
   - set wiper switch to stage 1
   - ignition off
   - ignition on
   - ignition off
   - remove key from ignition lock
   - insert key into ignition lock and turn to pos. I **or** to buzzer contact (not "**ignition on**"). If the key is in the correct position, either the wipers must be switched on in pos. I or, in buzzer contact position, the test lamp or selector LED must light up (see examples on page 2).
   - Remove key from ignition lock.

7. After the key has been removed, you have **a total time of 1 minute for tuning** the remote control units. During this 1 minute, you must tune one remote control unit after the other to the vehicle by pressing the pushbutton. When tuning the units, you must point the unit with the button pressed towards the control unit for **at least 5 seconds**.

8. The status "tuning of remote control units" is ended when:
   - 4 remote control units have been tuned
     or
   - a fifth remote control unit is operated within the minute
     or
   - one unit is operated twice
     or
   - the minute has elapsed.

Note

Any remote control units already assigned to a vehicle which are not tuned during a tuning session lose their validity for the vehicle.

Emergency unlocking of the door using the key

Note

The key should only be used for emergency unlocking in the event of a failure of the remote control unit.

1. Hold the key in the unlocking position and open the door at the same time using the door handle. The alarm is triggered immediately.

2. It is not possible to deactuate the immobilizer or the alarm system. The engine cannot be started.

3. The door is locked again as soon as it is closed. The central locking system remains in operation.

90 68 Immobilizer from model '95

Emergency unlocking of the door using the key

The key should only be used for emergency unlocking in the event of a failure of the remote control unit.

De-activation of immobilizer (with vehicle locked)

It is possible to switch off the acoustic alarm for 2 minutes so that the immobilizer may be de-activated by **entering a four-digit code number specific to the vehicle**. The four-digit code number (key card) is entered using the ignition key by a **sequence of ignition on and off operations**. The time available for entering the number is **100 seconds.** Each digit of the code number is determined by the number of **ignition OFF-ON** operations, starting with the left digit. To enter "0", the ignition must be switched from off to on 10 times.

You must **wait for the warning light to go on before entering the next digit**.

1. The door lock must be **unlocked, locked and unlocked again in 5 sec.** The acoustic alarm is then interrupted after 1 second.

2. Open the door within **10 seconds** and switch the **ignition** on. The **warning light** in the clock will then light up and go out again after **15 seconds.**

3. **Switch the ignition off and on again within 5 seconds. The warning light** will light up and start to **flash after 15 seconds.**

4. You must start to **enter the code number** within **5 seconds**.

5. Example: code number 1312

1 = ignition off-on
wait for warning light
3 = ignition off-on, off-on, off-on
wait for warning light
1 = ignition off-on
wait for warning light
2 = ignition off-on, off-on

6. If the number is correct, **the warning light will flash.**
   This is also the point at which tuning of the remote control units can start.

Note

After the warning light has flashed (point 6) and the ignition has been switched on again, the engine can be started.

The immobilizer is deactivated and the alarm system and central locking system are unlocked. If the warning light does not flash after the code number has been entered, steps 1-5 must be repeated.
Emergency deactivation does not deactivate the self-priming function of the immobilizer system.

Printed in Germany - 13, 1995

Tuning remote control units

Note

A maximum of four remote control units may be assigned to a vehicle at the same time. Additional remote control units must **always** be tuned to a vehicle **together** with the two units already assigned to it. Before the code number is entered, the **immobilizer** must be primed but the alarm and central locking systems must not be activated. This status is reached **90 seconds** after the ignition key has been removed or **3 minutes** after the vehicle has been unlocked using the remote control unit if the ignition key is not turned in the lock.

Entry of code number

1. Switch the **ignition** on. The **warning light** in the clock will go on and off again after **15 seconds**.

2. Follow the instructions in points 3 to 6 on page 7.

3. If you have reached the tuning status (point 6 page 7 **warning light is flashing**) the remote control units **must be tuned within one minute.**

4. Press the pushbuttons of the remote control units until the **LEDs** of the alarm system in the doors **flash** (as an acknowledgement). Operate all the remote control units one after the other.

5. If no more remote control units are operated within **1 minute** or the ignition is switched off, tuning is stopped.

911 Carerra RS vehicles

Note

As Carrera RS vehicles are not equipped with a central locking system, both doors must be locked and unlocked using the key.

To deactivate the immobilizer, the remote control unit must be **operated twice** in the following situation.
The immobilizer has self-primed and the ignition has not been switched on.

If the ignition is switched on before operating the remote control unit or the remote control unit is used to activate the immobilizer, the remote control unit must only be operated **once.**

The procedure for tuning remote control units is the same as for the 911 Carrera (993) from model '95.

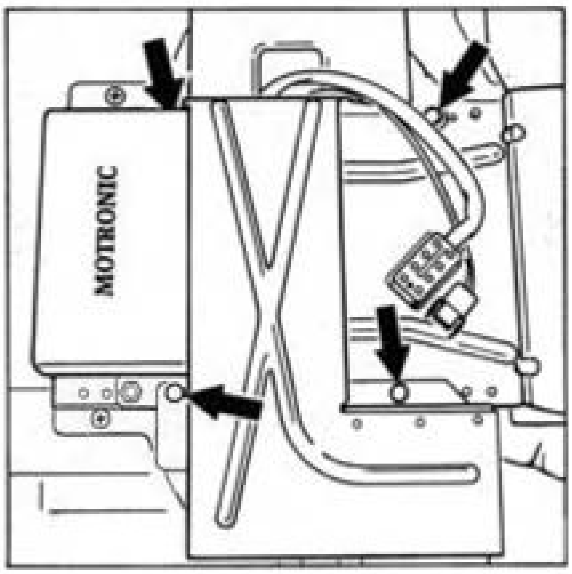
Printed in Germany - I3, 1995

90 68 19 Removing and installing immobilizer control unit

Removal

1. Remove left seat.

2. Remove shear bolts using a cross cut chisel and remove cover.



2006-24

3. Loosen control unit and unplug connector.

Installation

Break off new M 6 x 13
shear bolts, part no. 999 074 051 02 using a Torx E6 socket wrench.

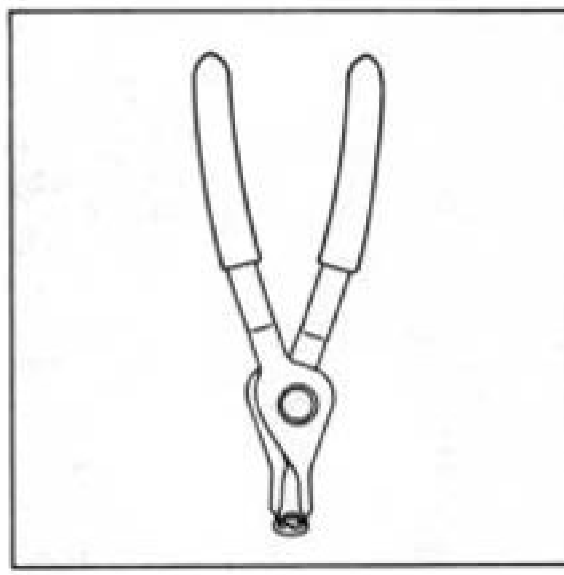
Removing and installing control unit of RS vehicles

Removal

Loosen flanged nut using a cross cut chisel.

Installation

Tighten new M5 flanged bolt with stop teeth (part no. 999 507 472 02) using circlip pliers, e.g. Hazet 4843a-12.



2219-24

Note

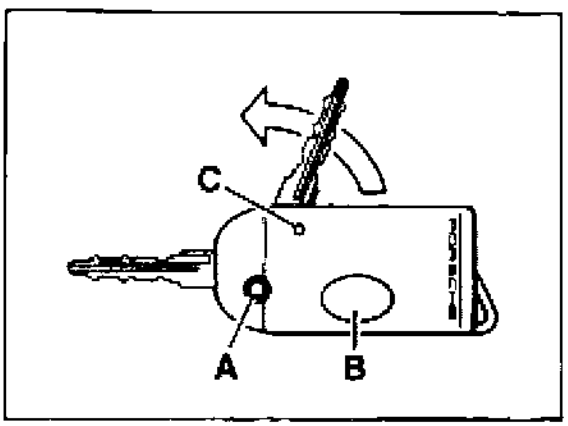
Tighten the control unit carefully. The plastic mounting lugs can break off. The serial number must **not be left on the control unit** after installation. The two stickers must be handled confidentially together with the customer. One sticker must be attached to the invoice and the second sticker given to the customer.

90 73 37 Disassembling and assembling hand-held transmitter

Car keys with integrated hand-held transmitter for vehicles with immobilizer

Note

The car key has been designed as a folding key.
Folding out:
Hold housing in your left hand and press "A" button.
Folding in:
Press "A" button and fold key into the housing manually.



2307-90

A - Button to fold the key in and out

B - Key for remote control

C - LED

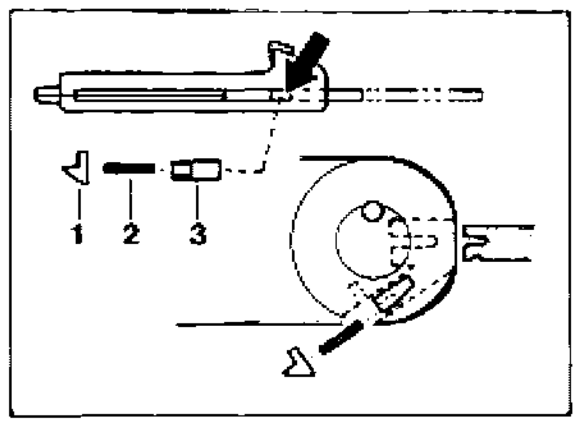
Disassembling housing with key

1. Slacken screw in upper part of the housing and remove upper part.

2. Carefully remove battery and printed-circuit board.

3. Fold out car key with "A" button from lower part of the housing.

4. Carefully open and pull out cap using a small, pointed screwdriver and working from the front (arrow). Remove compression spring.

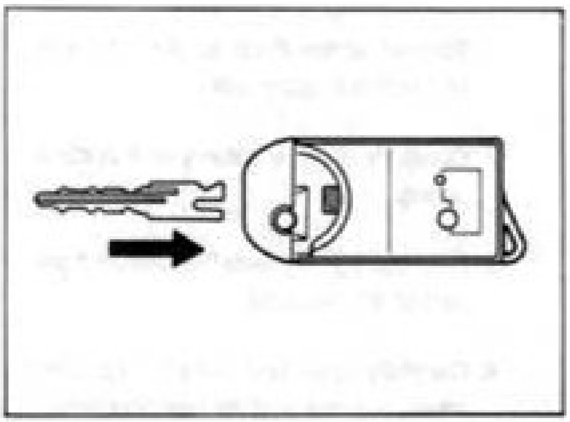


2308-90

1 - Cap
2 - Compression spring
3 - Locking slide

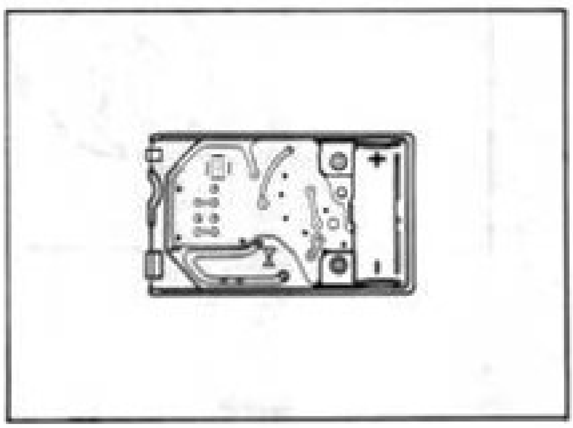
5. Push locking slide in the lower housing part slightly back using a screwdriver and pull out key.

6. Reinstall compression spring and cap. Snap new key into the lower part of the housing.



2310-90

7. Install printed-circuit board in correct position in the upper part of the housing. Install battery (observe polarity). Plus and minus are marked in the upper housing part.

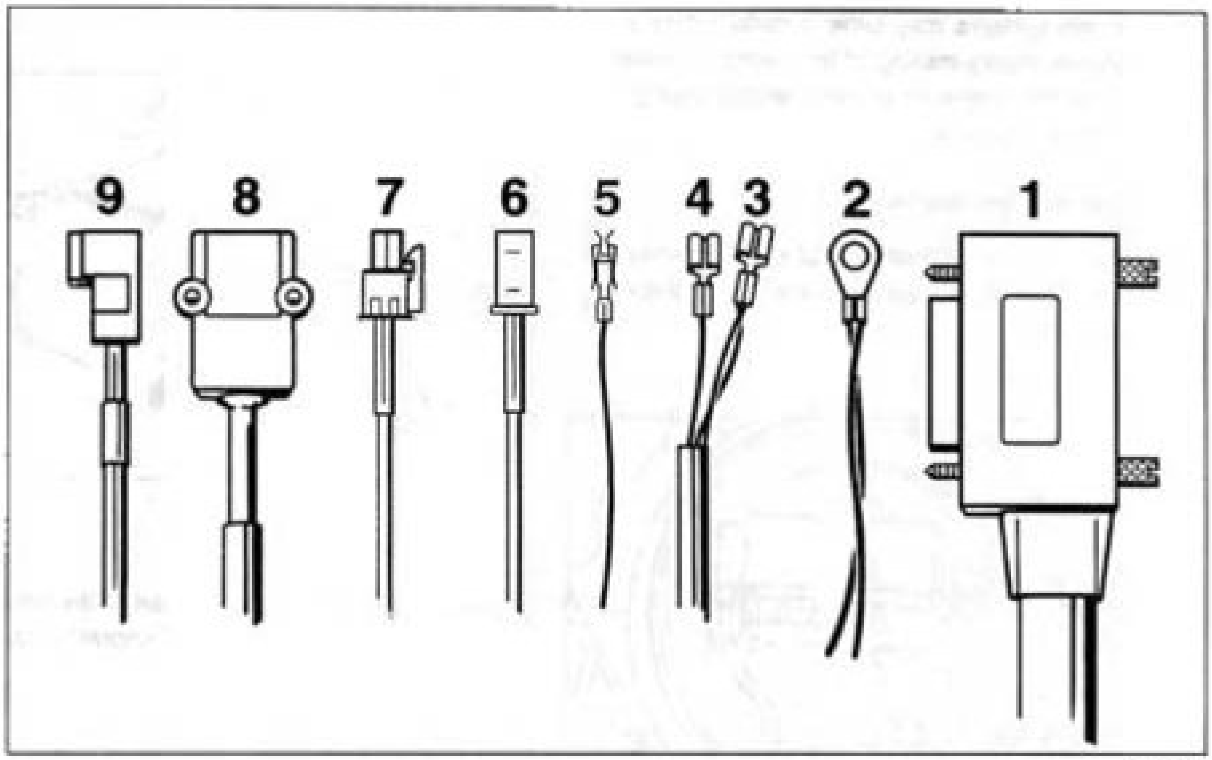


2311-90

8. Fit upper housing part and check operation.

91 55 23 Retrofitting a D-Net telephone system

D-Net telephone wiring harness



1962-91

1 - Transmitter/receiver connection

2 - Ground / ground point VI

3 - Term. 30 connection

4 - Term. 15 connection

5 - Radio muting connector III - black
Contact No. 2

6 - Microphone connection

7 - Speaker connection

8 - Separate card reader connection

9 - Operating unit connection

Rubber grommet and terminal are fitted to the wiring harness.

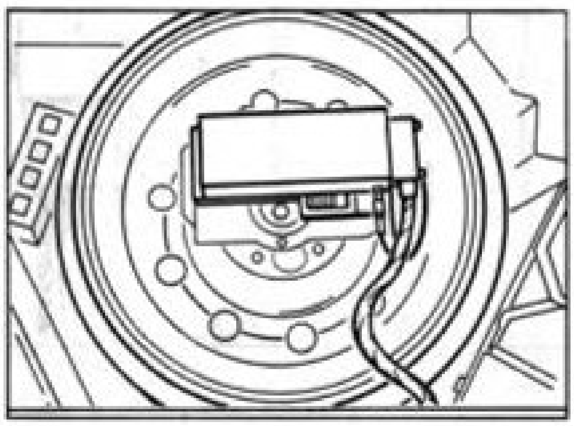
Installation position of components

Note

Retrofitting is described taking the Motorola International 2000 as an example. Other telephone systems may differ in certain details. Do not modify routing of the wiring, however, since this may lead to malfunctions during vehicle operation.

Transmitter/receiver unit

The transmitter/receiver unit is fitted within the emergency spare wheel in the luggage compartment.

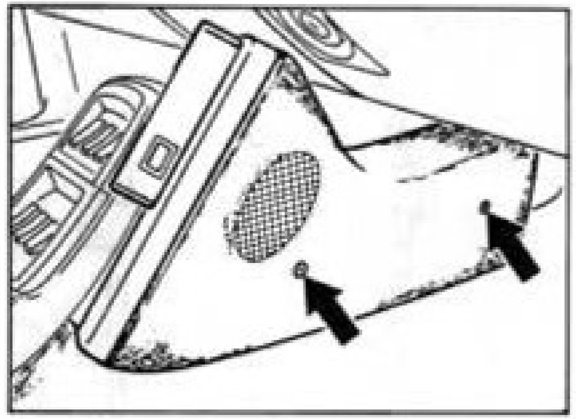


1950-91

The mounting bracket for the transmitter/receiver unit is included with the respective telephone version.

Telephone console

The telephone console is bolted to the right-hand side of the center console in the footwell.

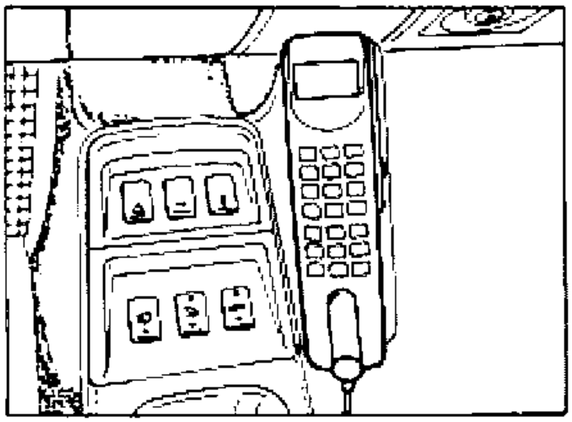


1951-91

Fit seat stop for passenger seat to the left seat rail below the front hexagon socket head bolt.

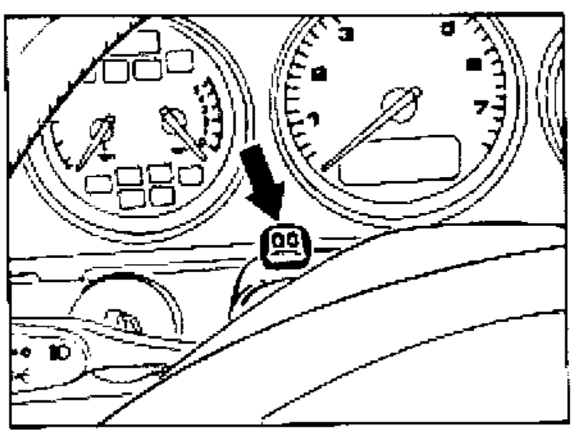
Operating unit

Assembly instructions and/or drilling template are supplied with the respective telephone model.



Handsfree microphone

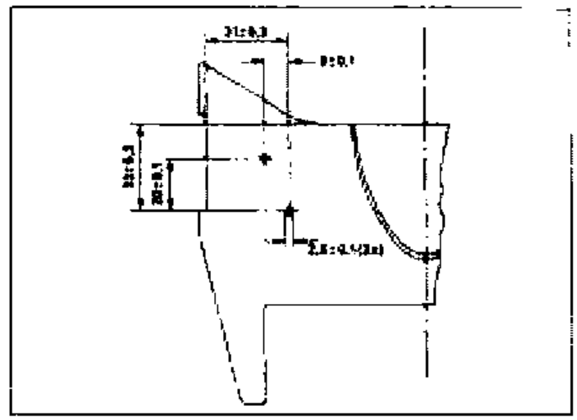
The handsfree microphone is located in the upper section of the steering column panel.



1953-91

Hole pattern for handsfree microphone assembly panel.

The figure below shows the hole pattern from below (upper section of steering column panel).



1954-91

Fitting an antenna to the Coupé

Center of vehicle and **100 mm** from rear edge of roof panel.

Drill hole dia. 18 mm

Fitting an antenna to the Cabriolet

Vehicle center and center of windshield outer frame/soft top mounting.

Drill hole dia. 18 mm

Note

The windshield antenna must **only** be used for radio operation.

Routing the wiring

Always insert suitable rubber grommets when routing wires through drill holes in sheetmetal panels.

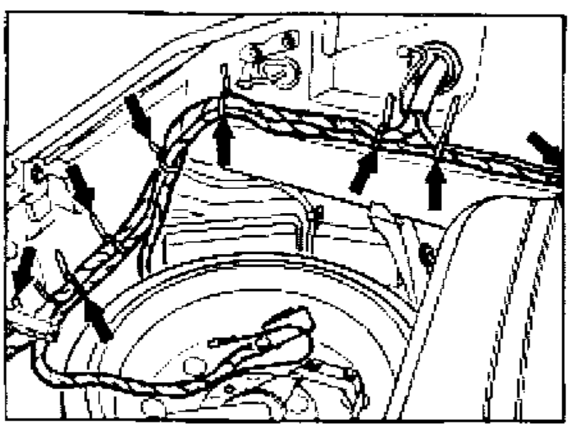
Antenna cable

The antenna cable is routed from the passenger footwell across the right-hand A-post along the roof panel towards the B-post and to the antenna exit. Make sure the antenna cable does not cause any clattering noise.

The sun visor and courtesy light will have to be removed for fitting of the antenna cable.

D-Net wiring harness

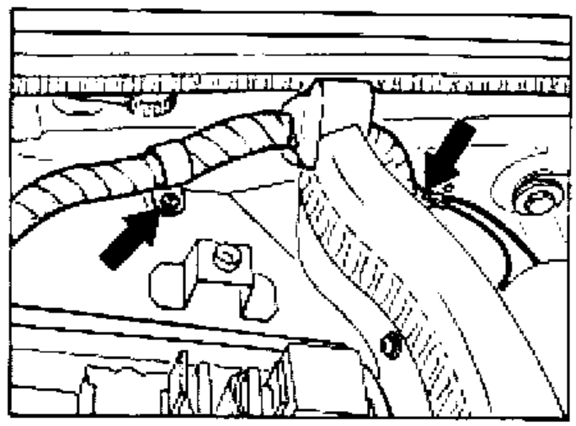
The wiring is routed from the transmitter/receiver unit towards the front wiring harness. It is tied to the front-end wiring harness with standard tie-wraps. In the strut area, a mark to aid assembly is attached to the wiring harness.



1955-91

Attach white mark to standard tie-wrap location.

Fit edge protector to strut tower area. Attach telephone wiring harness with wire clamp to wiring harness cover. Connect ground to ground point VI in the cover hinge area. Fit end connector to D-Net wiring harness in transverse wall.



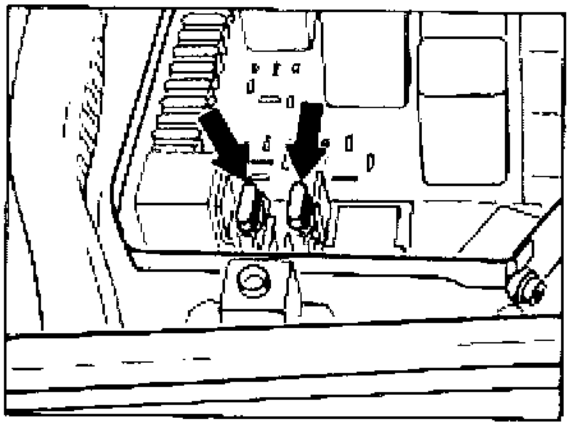
1956-91

Connect power supply wires (term. 15 and term. 30) to Central Electrical System.

Term. 30 is connected across a fuse to the screw-on terminal at the Central Electrical System.

Term. 15 is connected across a fuse to connector K 13.

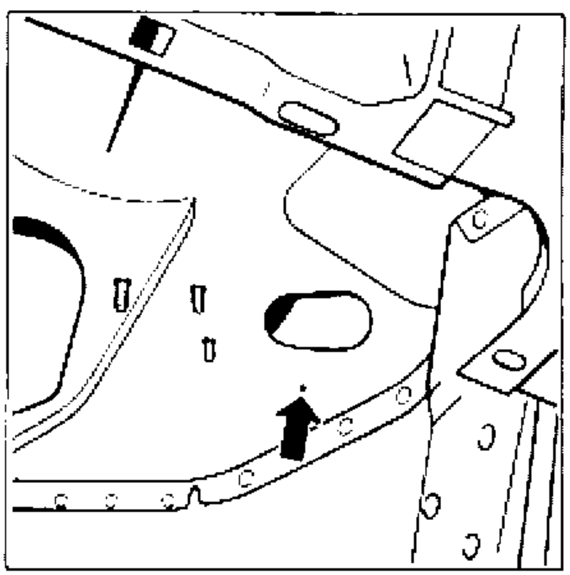
The fuses are installed in a fuse holder in relay slot R 62.



1957-91

Fusing of term. 15: 7.5 A
Fusing of term. 30: 7.5 A

The wiring harness is ruoted across a separate 41 mm dia. drill hole into the passenger compartment. Make drill hole at the indicated centerpunch point in the Transverse wall. The rubber grommet destined for this purpose is fitted to the wiring harness



802-91

The power supply leads to the operating unit and handsfree microphone are routed along the passenger compartment wiring harness to the tunnel and along the tunnel upper edge.

Note

On **M 692** (optional radio with **CD changer**), the radio muting wire no. 5 (telephone mute line) is **insulated** and tied to the harness.

Exception:

With the München RD 104 (M 687) radio **with sound package** (M 490), the telephone mute line must be connected to contact 4 of connector I.

If a sound package is **not** installed, the telephone mute line must not be connected and must remain insulated.

With the Düsseldorf RCR 84 (M336) radio, the telephone mute line must not be connected: tie up the **insulated** line.

See also technical information bulletin no 70/94 for 911 serie.

Printed in Germany - 12, 1995

91 20 01 Check list of possible faults on the Porsche Autoradio CR 10

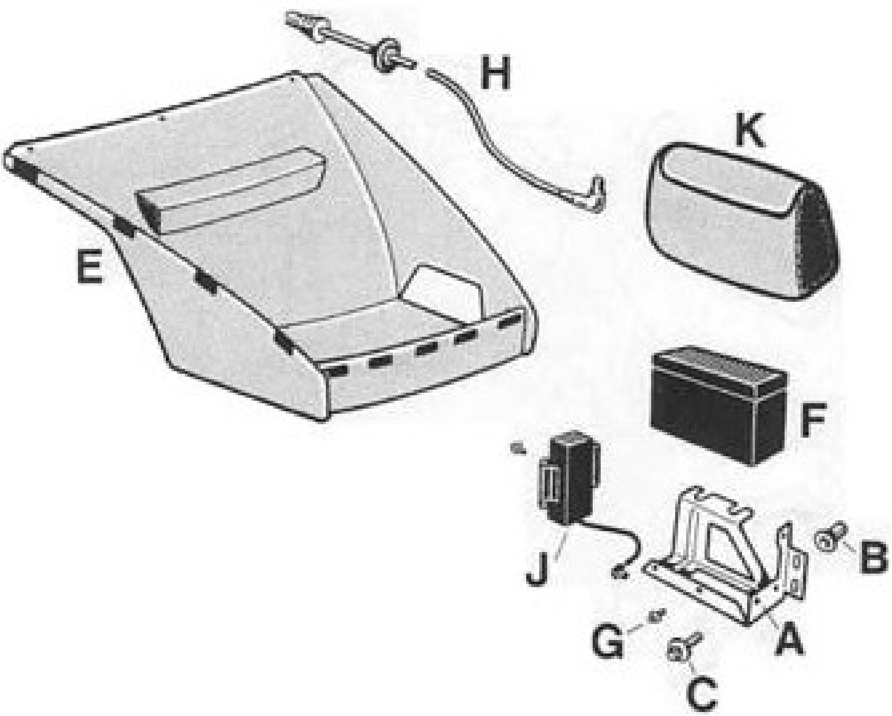
Symptom	Reason	Remedy	Refer to radio Op. Manual page:
• Radio cannot be switched on	• Fuse faulty	• Replace fuse	53
	• Incorrect polarity of wiring harness wires ⇨ Faulty fuse	• Replace fuse, Remove fault in harness	
	• Pin in connector block chamber A is bent	• Realign	53
• Unit switches off by itself	• According to specification; After 1 h, if terminal 75 (ignition) off	• Switch unit on again	
• Code input not possible	• Incorrect or no code card	• Compare series No. on unit and code card ⇨ Inform Becker GmbH company	32 + 33
• "WAIT" on LCD	• Incorrect code was entered 3 times	• **Caution:** Unit must remain on for 10 minutes until new code can be entered	32 + 33
	• Incorrect code was entered 9 times	• **Caution:** Unit must remain switched on for 60 minutes until new code can be entered	32 + 33
• No sound; Low sound; Sound different at front, rear, left or right	• Fader or balance settings not in center position	• Adjust fader or balance	34 + 35
	• " Phone" display on LCD	• Mute via telephone?	46
• Fader, balance cannot be adjusted	• Active traffic program	• Wait for end of traffic program	44 - 46
• Optical theft protection does not flash	• Function is switched off	• Function can be adjusted	33
	• LED faulty	• Inform Becker GmbH company	33

Symptom	Reason	Remedy	Refer to radio Op. Manual page:
• Alarm is triggered constantly	• Insulating plate on chassis missing	• Replace radio	
	• Mounting frame rotated by 180°	• Rotate mounting frame by 180°	
• Alarm inoperative	• Wire not connected	• Connect wire	
	• Contact on mounting frame bent	• Straighten	
• No sound	• Speaker chamber B not connected, Connector not connected properly	• Connect unit correctly	53
	• Sound system chamber C not connected Connector not connected correctly	• Connect unit correctly	53
	• Pins in connector block of chamber B or chamber C bent	• Straighten	53
	• Speaker faulty	• Check speaker	
	• Wiring harness missing, incorrect or faulty	• Check wiring harness	53
	• Volume adjustment	• Increase volume adjustment	34
	• Sound system faulty	• Check sound system	53
	• Unit is muted	• Deactivate muting	45
• Sound present in spite of muting	• Only during initial 20 seconds after turning on if muting was activated when unit was last turned off	• None, sound is automatically switched off after 20 sec	45
• Cassette winds automatically during CC operation	• According to specifications (in blank sections) ⇨ Skip-blank function	• Deactivate Skip-blank function	48
• "CLEAN" displayed during CC operation and beep of approx. 1 sec.	• After approx. 30 hours of cassette operation ⇨ User display	• Use cleaning cassette	49

Symptom	Reason	Remedy	Refer to radio Op. Manual page:
• No sound in CD operating mode	• Pins in connector block chamber C bent	• Straighten	53
	• Wiring harness for CD changer missing, incorrect or faulty	• Check wiring harness	53
• No radio reception, noise, no station names	• Antenna cable not connected or faulty	• Check antenna cable	53
	• Antenna not raised (mech.)	• Pull out antenna	
	• Antenna not raised (electr.); Control lead not o.k., Pin 5 of chamber A bent	• Connect control wire; Check control wire. Straighten	53
• No station names are displayed	• According to spec. ⇒ Fix mode	• Switch to RDS mode	38 + 39
	• Poor reception quality, No RDS sensing possible	• Tune in to other station	38 + 39
• Tuned station is no longer on station button	• Caution. 3 levels: FM mode, TP mode, Fix mode	• Adjust to correct level	37 - 39, 42, 44
	• Poor reception quality, no RDS sensing possible, "- - - - - - - - - -" in display	• Tune in to other station	15
• Sound adjustments have changed	• When radio is reconnected, automatic basic setting is activated	• None	
• Sound adjustments differ	• 2 user sound memories exist	• None	36
	• Different sound adjustments for FM, traffic program, AM, CC and CD modes	• None	34
• Muting no longer turned on	• If no traffic stations can be received, muting is automatically deactivated	• None	19
• Radio shifts automatically between stations	• Only for regional stations: Regional operation is off ("RP OFF")	• Regional operation must be switched on ("RP ON")	43

91 60 23 Retrofitting of a CD changer

The following spare parts are required for retrofitting of a CD changer:



1154 - 91

A = Bracket for CD changer

B = Pop rivet nut M 6/0.9 - 3

C = Combination screw M 6 x 16

E = Front compartment carpet, modified for CD changer

F = CD changer

G = Combination screw M 4 x 8

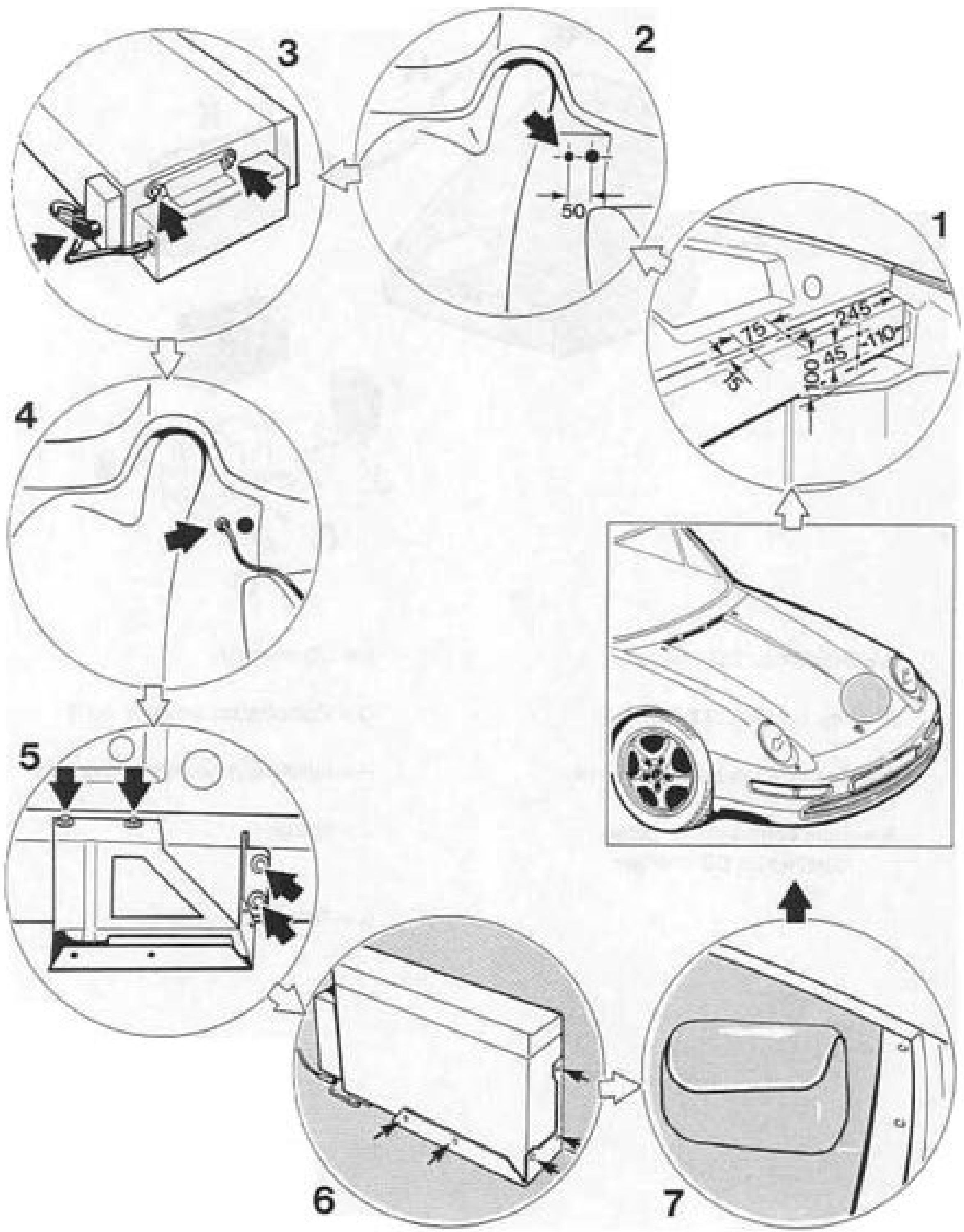
H = Interface/radio connecting wire

J = Interface

K = Trim for CD changer

Printed in Germany - 12, 1995

Retrofitting of a CD changer



Printed in Germany - 12, 1995

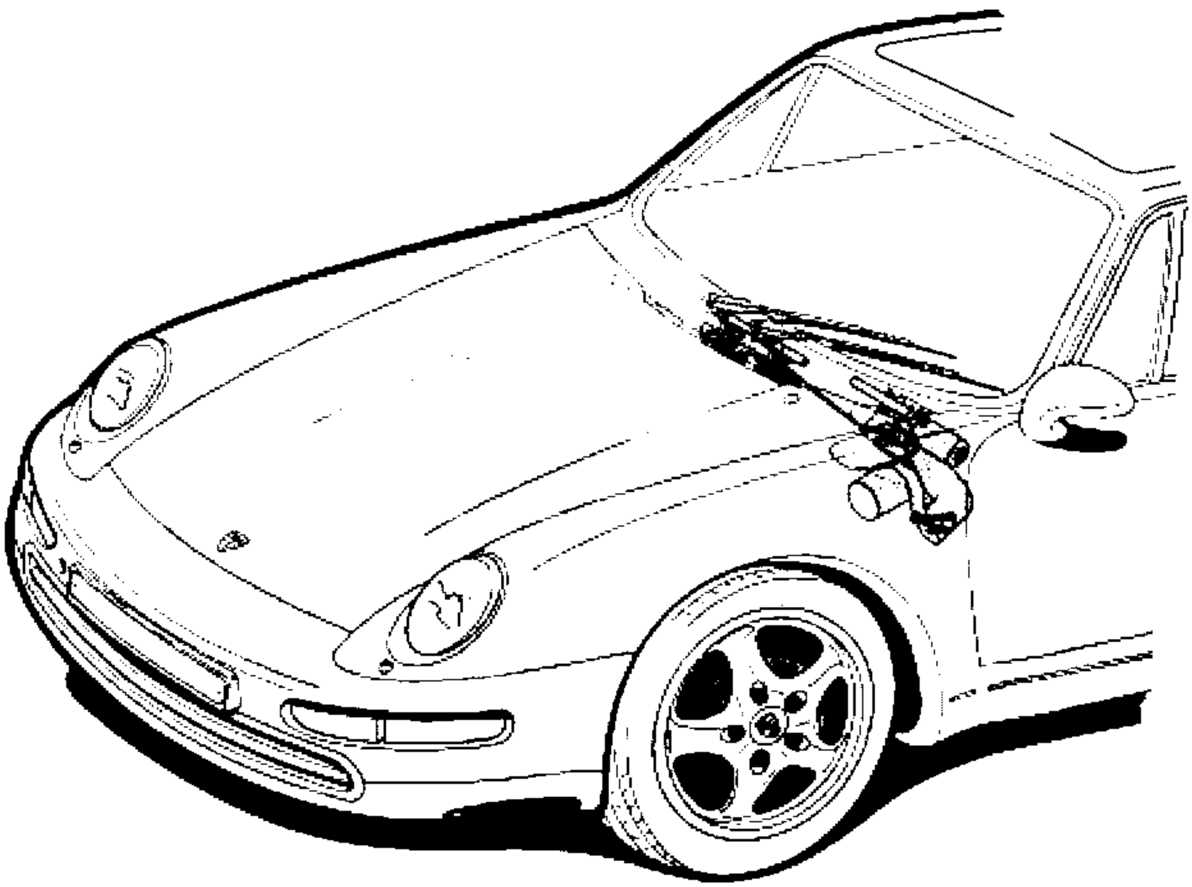
Retrofitting a CD changer

Remove front compartment carpet, spare tire, protective cover, fuel tank and battery!

No.	Operation	Instructions
1	Insert pop rivet nuts for attachment of the CD changer bracket into side member	Scribe attachment points for CD changer bracket on left-hand front end of side member. Using a 9.5 mm dia. drill bit, drill holes into side member according to scribe marks. Insert M 6 pop rivet nuts into side member and rivet in place using a pop rivet nut gun,
2	Drill 20 mm dia. hole for routing of connecting wire	Scribe hole for routing of CD changer/radio connecting wire on bulkhead and drill to 20 mm dia.
3	Finish assembly of CD changer	Remove transport locks from CD changer. Cover open holes on CD changer using the self-adhesive sheets supplied. Fit interface to CD changer using M 4 x 8 combination screws (4 pc.). Connect interface/CD changer wiring.
4	Insert connecting wire	Route connecting wire across hole in bulkhead, put grommet into place and insert connecting wire into radio. Tie connecting wire to wiring harness along side member.
	Install and connect battery	Enter radio code, correct clock setting and carry out system adaptation.
5	Fit bracket for CD changer	Fit CD changer bracket to side member, using M 6 x 16 combination screws (4 pc.).
6	Fit CD changer	Plug connecting wire into interface (CD changer). Fit CD changer to CD changer bracket, using M 4 x 8 combination screws (5 pc.).
	Check CD changer operation	
7	Fit spare tire, modified front compartment carpet and trim for CD changer	

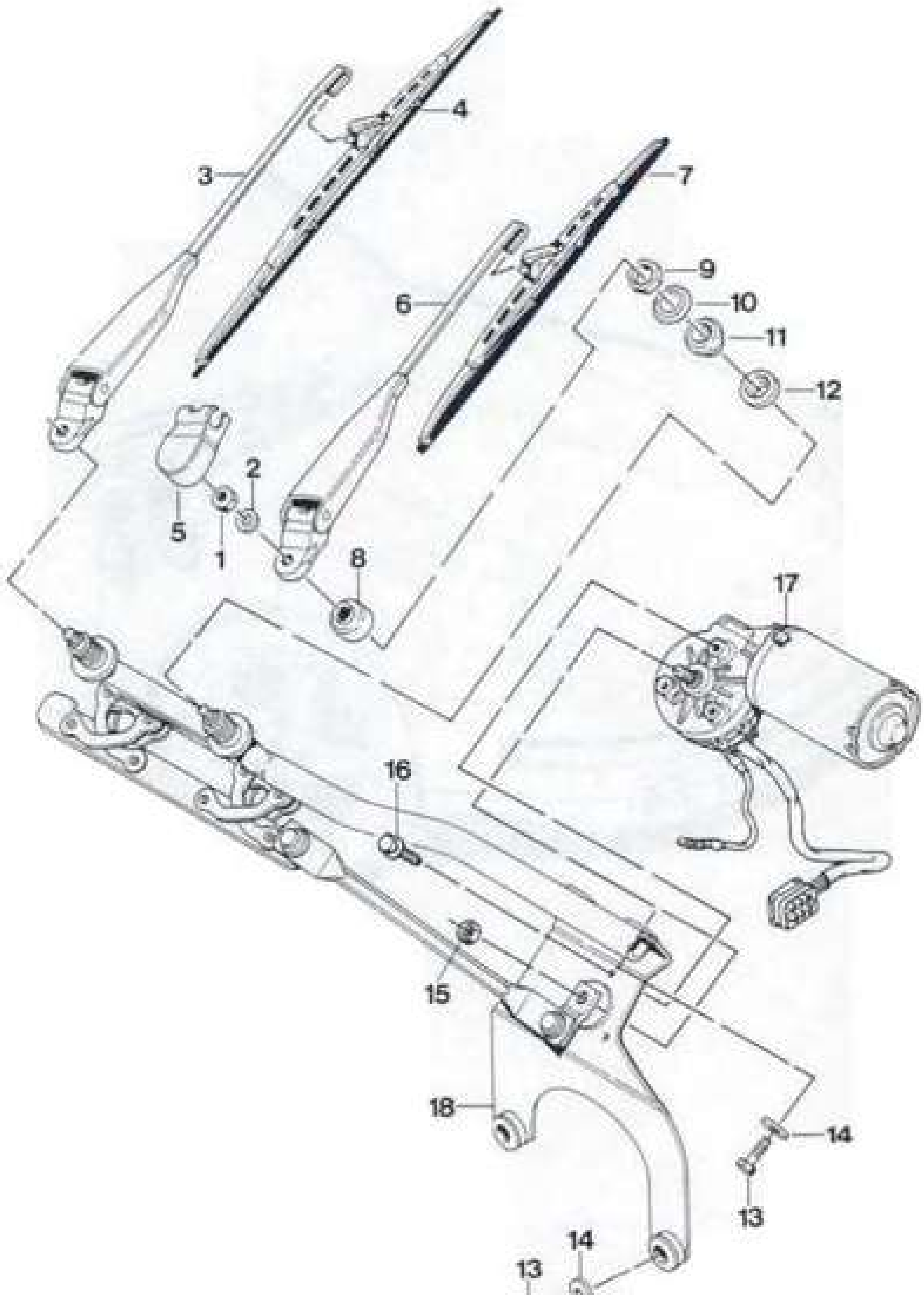
Printed in Germany - 12, 1995

92 00 19 Removing and installing windshield wiper system

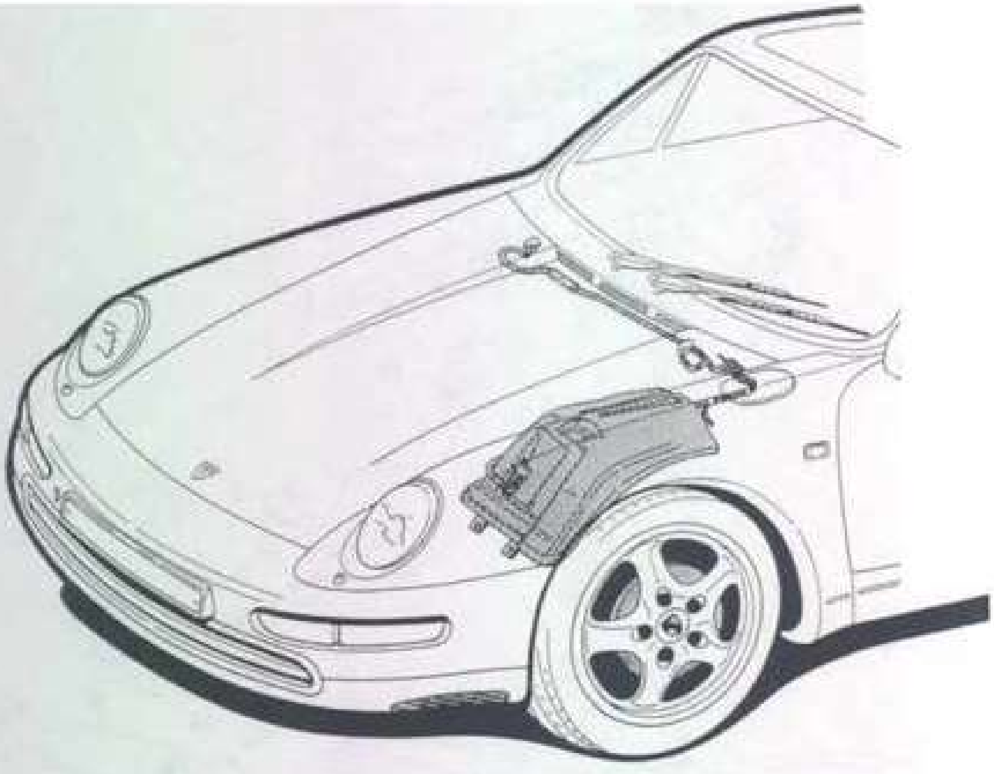


1625 - 92

92 00 19 Removing and installing windshield wiper system

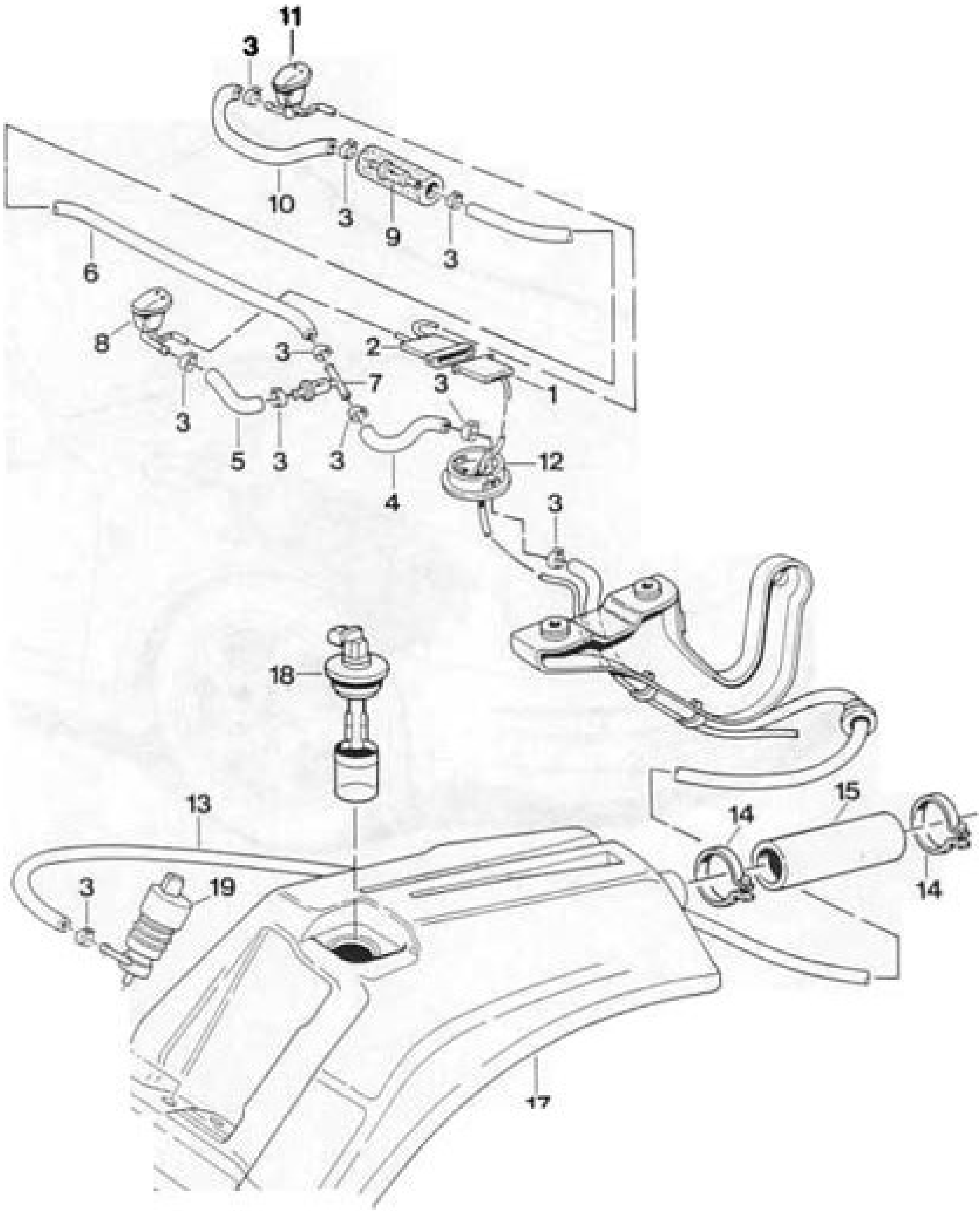


No.	Designation	Qty.	Note: Removal	Installation
1	Hexagon head nut M 8	2		MA = 14 - 16 Nm (10 - 12 ftlb.)
2	Washer	2		
3	Wiper arm right-hand	1		
4	Wiper blade right-hand	1		
5	Cap	2		
6	Wiper arm left-hand	1		
7	Wiper blade left-hand	1		
8	Cap	2		
9	Hexagon head nut M 18 x 1	2		MA = 10 +1 Nm (7 + 0.7 ftlb.)
10	Flange washer	2		
11	Rubber bushing	2		
12	Rubber bushing	2		
13	Hexagon head screw M 6 x 22	4		MA = 10 Nm (7 ftlb..)
14	Washer A 6.4	4		
15	Nut	1		
16	Hexagon head screw	3		MA = 11 Nm (8 ftlb.)
17	Wiper motor	1		
18	Crank gear	1		



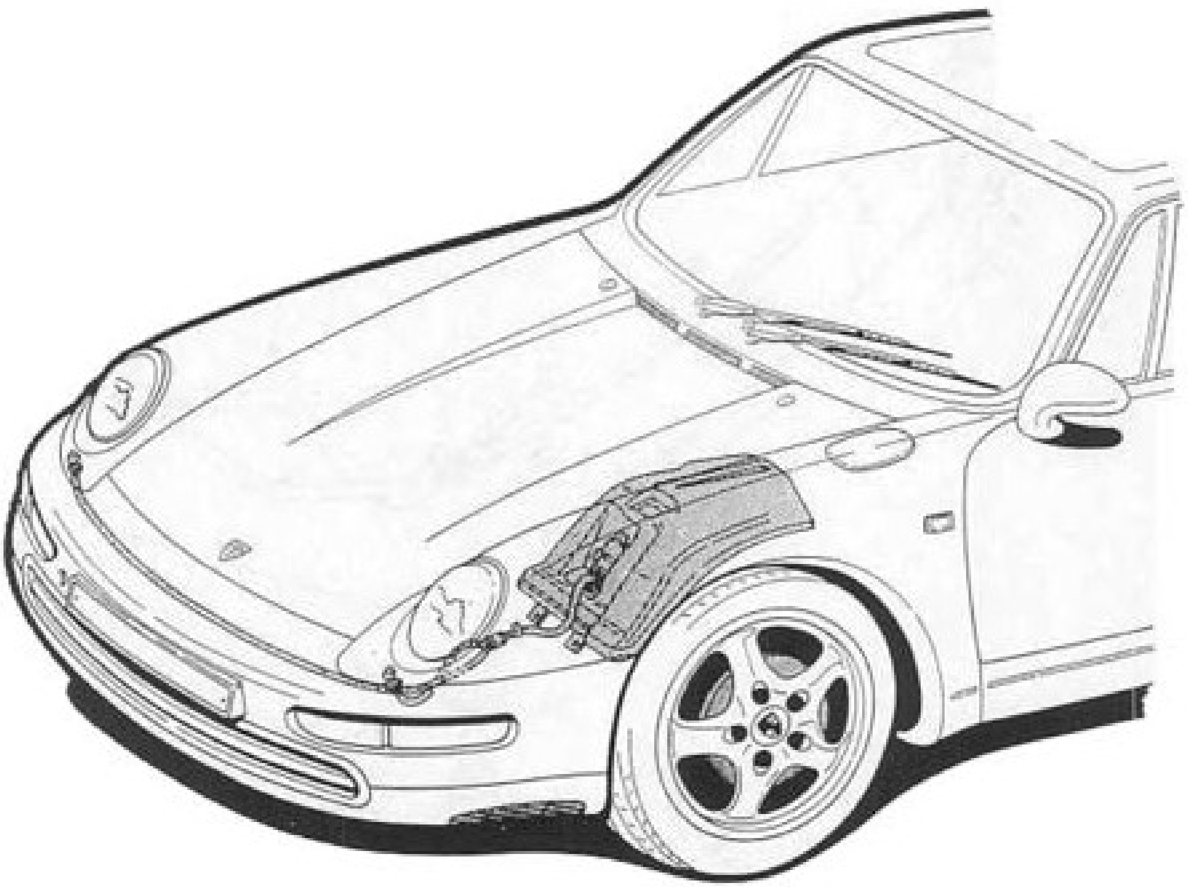
92 00 19 Removing and installing windshield wiper system

92 00 19 Removing and installing windshield washer



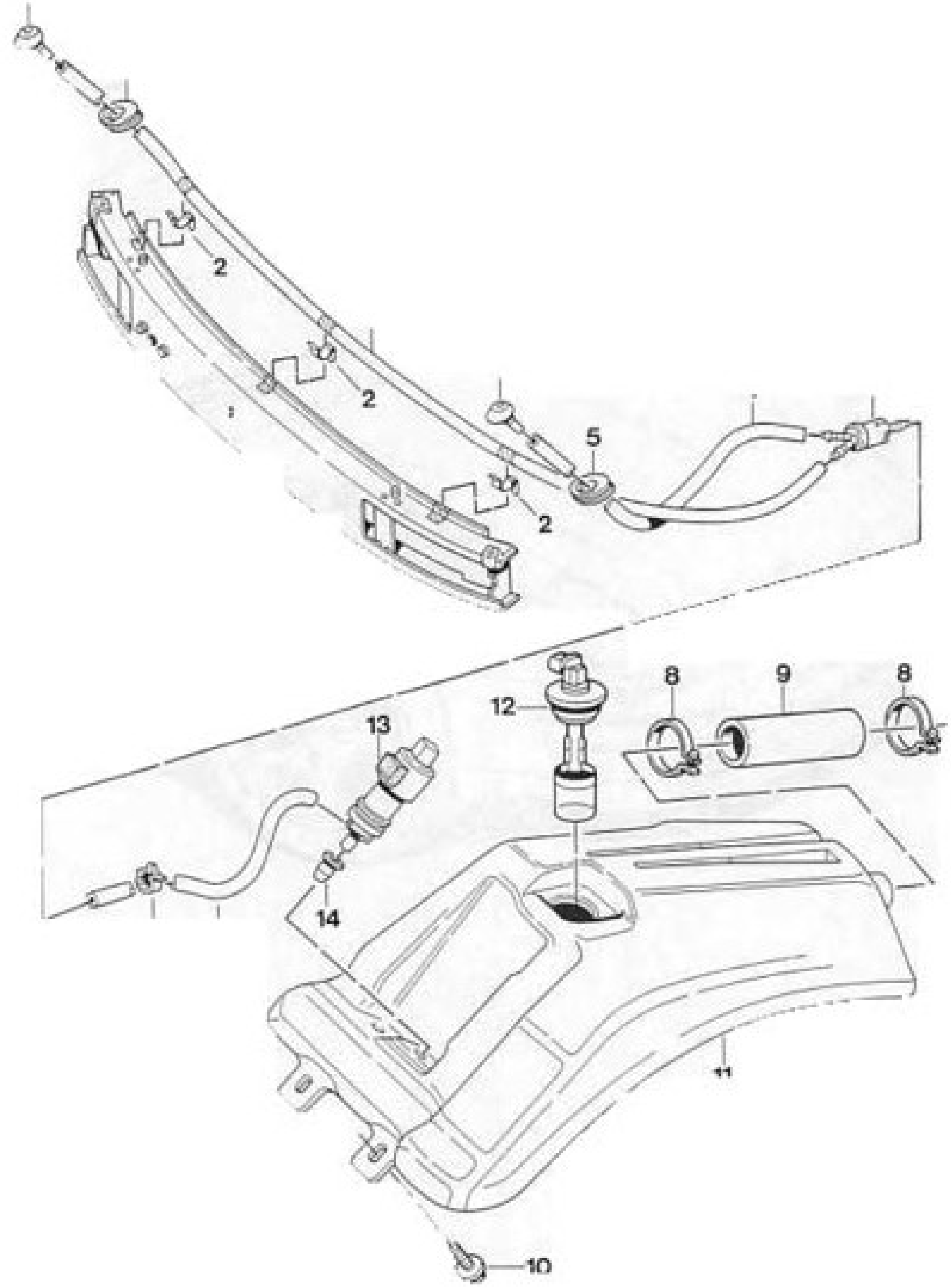
No.	Designation	Qty.	Removal	Note: Installation
1	Passenger wiring harness connector	1		
2	Connector housing for heated spray nozzle	1	Disengage wires with commercially available extracting tool	
3	Ear clamp	10		Replace
4	Hose	1		
5	Hose	1		
6	Hose	1		
7	Valve	1		
8	Spray nozzle left-hand	1	Push back and disengage lug	
9	Valve	1		
10	Hose	1		
11	Nozzle right-hand	1	Refer to No. 8	
12	Hose clamp	1		
13	Hose	1		
14	Hose clamp	2		
15	Hose	1		
16	Screw	2		
17	Water tank	1		
18	Sender for fluid level indicator	1		
19	Pump for windshield washer system	1		
20	Gasket	1		

92 00 19 Removing and installing headlight washer system



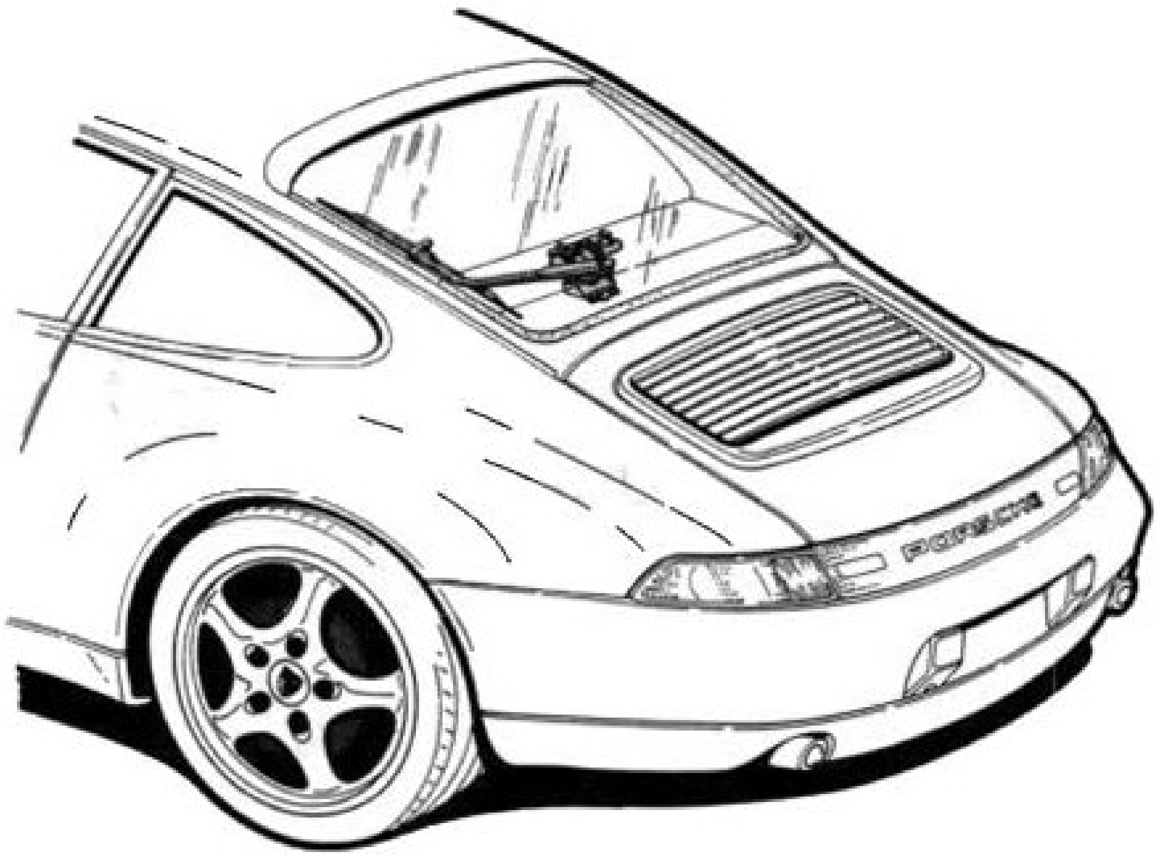
1827 - 92

92 00 Rem and insta ng headlight washer system



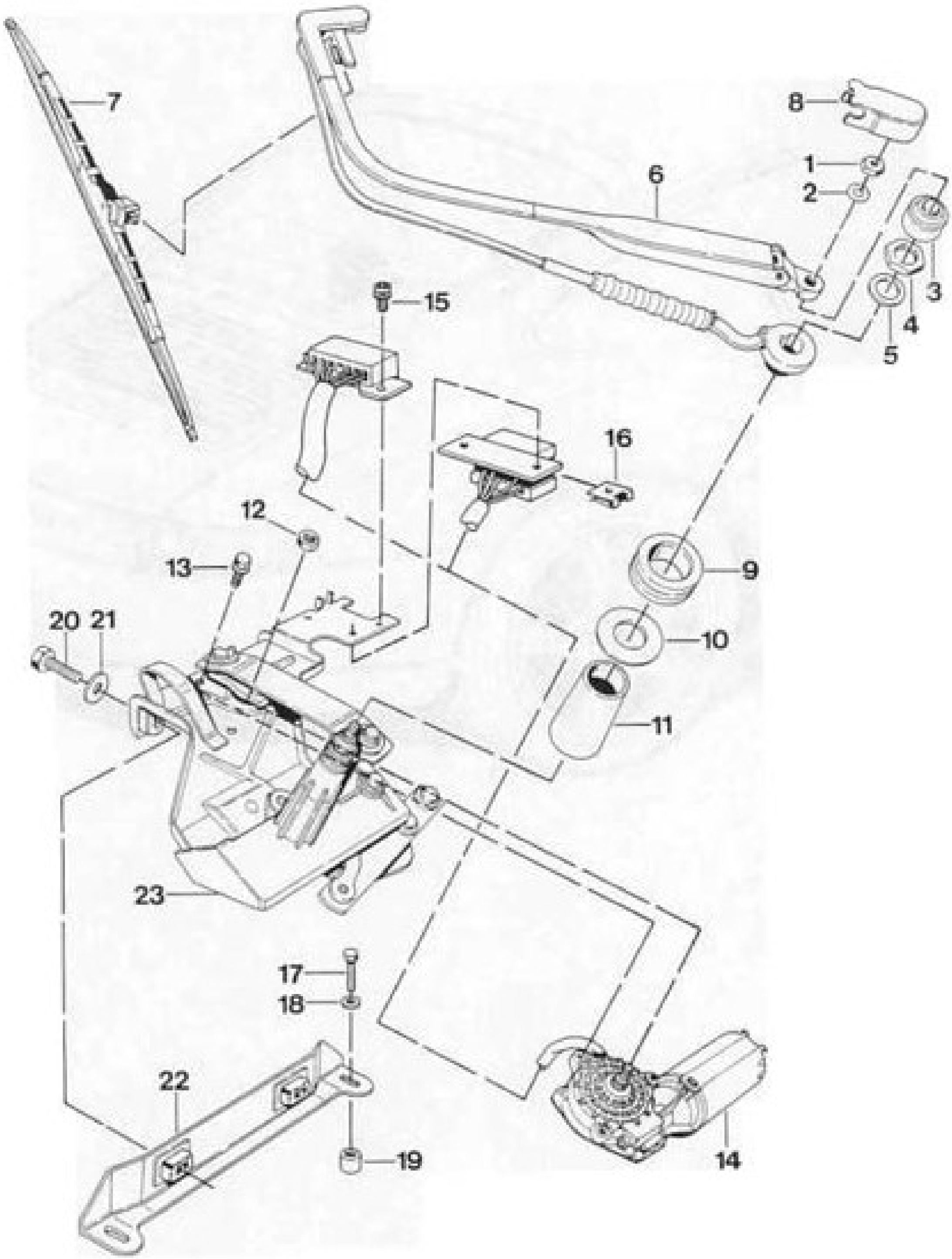
No.	Desigantion	Qty.	Note: Removal	Installation
1	Spray nozzle	2		
2	Clamp	3		
3	Hose	1		
4	Hose	2		
5	Grommet	2		
6	Bracket	4		
7	Valve	1		
8	Refer to p. 92 - 7 No. 14			
9	Refer to p. 92 - 7 No. 15			
10	Refer to p. 92 - 7 No. 16			
11	Refer to p. 92 - 7 No. 17			
12	Refer to p. 92 - 7 No. 18			
13	Pump for headlight washer system			
14	Gasket			

92 00 19 Removing and installing rear window wiper



1629 - 92

92 00 19 Removing and installing rear window wiper



No.	Designation	Qty.	Note: Removal	Installation
1	Nut M 8	1		M_A = 20 - 25 Nm (15 - 18 ftlb.)
2	Washer	1		
3	Cap	1		
4	Nut M 16 x 1	1		M_A = 14 - 16 Nm (10 - 12 ftlb.)
5	Washer	1		
6	Wiper arm	1		
7	Wiper blade	1		
8	Cap	1		
9	Rubber bushing	1		
10	Washer	1		
11	Cover	1		
12	Nut	1		M_A = 10 Nm (7 ftlb.)
13	Screw	3		M_A = 11 - 12 Nm (8 - 9 ftlb.)
14	Wiper motor	1		
15	Self-tapping screw	3		
16	Sheetmetal nut	2		
17	Screw	2		M_A = 5 Nm (4 ftlb.)
18	Washer	2		
19	Bushing	2		
20	Screw	2		M_A = 8 - 10 Nm (6 - 7 ftlb.)
21	Washer	2		
22	Retaining bracket	1		
23	Assembly frame	1		

92 15 19 Removing and installing wiper motor (with A/C system)

Vehicles with air conditioning

Removal

1. Disconnect battery and cover battery or terminals.

Note

Follow safety precautions for work on vehicles with airbags.

2. Remove steering wheel and steering column panel.

3. Remove tachometer and large and small instrument cluster.

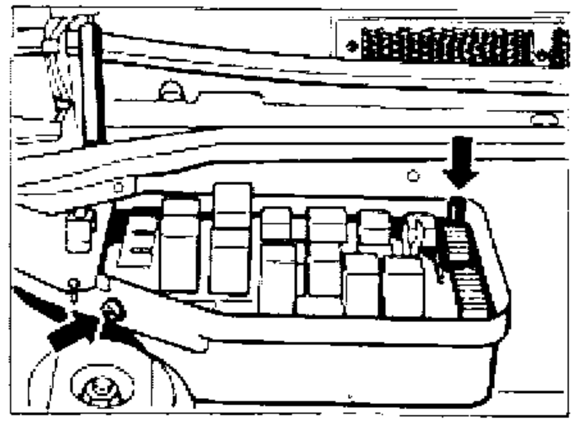
4. Remove air guide. Disconnect six-pin connector from wiper motor.

5. Unhook left pneumatic spring of luggage compartment lid and support luggage compartment lid.

6. Extract refrigerant with service equipment.

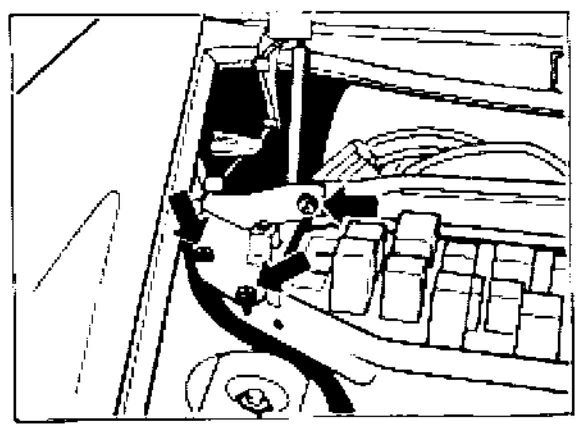
7. Remove cover of heater / A/C unit (old type with two screws and washers: new type with clamping spring).

8. Disconnect central electrical system.



240-87

9. Remove wire harness cover.



241-87

10. Unplug connector from lid of fan housing and lay central electrical system on fender (the fender must be protected by a cover).

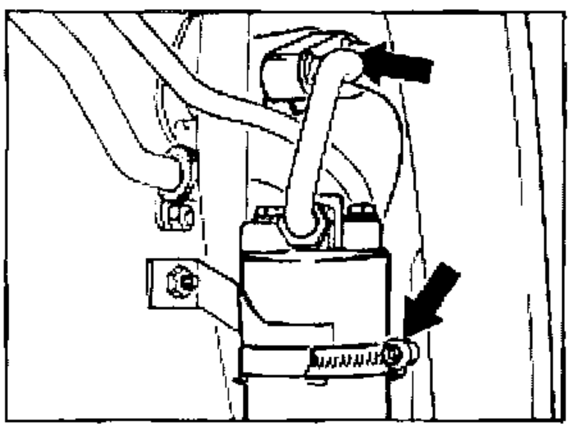
11. Unplug cable connector from final stage of fan.

12. Remove firewall.

13. Disconnect A/C lines from expansion valve. Immediately insert air-tight plugs in ends of all lines and connections. Loosen retaining clip.

14. Remove rear wheel housing liner on front left fender.

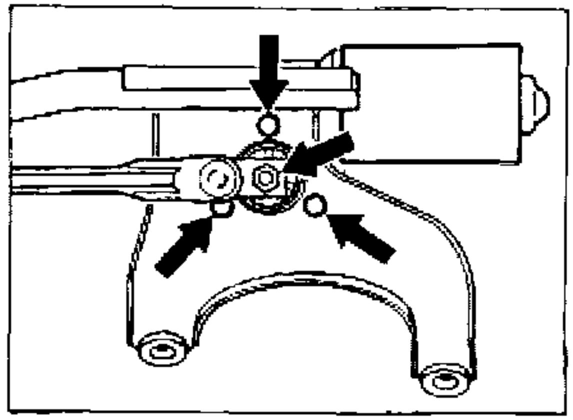
15. Disconnect hose clip from fluid reservoir. Undo mounting screw between wheel housing A/C lines and luggage compartment A/C lines. Immediately insert air-tight plugs in ends of all lines.



1897-87

16. Move A/C lines to side from wiper mount.

17. Release mounting nut of linkage, holding a 21 mm open-end wrench against linkage.



2072-92

Tightening torque: 22 ± 2 Nm (16 ± 1.5 ftlb.

18. Release mounting screws. Tightening torque: 11 Nm (8 ftlb.).

19. Remove wiper motor through installation opening of tachometer.

Installation

Note

Installation of the wiper motor is much easier if the motor is fixed in position with a stud bolt (sawn-off screw) and a nut. After fixing the motor, fasten it in place with 2 screws. Then unscrew the nut and stud bolt and insert the third screw.

1. Connect battery and switch wiper motor on and off (park position).

2. Install linkage. Cable and linkage must form a straight line.

3. Switch on wiper motor and check wiper positions.

4. **Disconnect battery again** and install the parts that you have removed. Use new O-rings and wet them with refrigerant oil.

5. Replace the two self-locking screws of the airbag unit in the steering wheel.

Note

To remove the crank mechanism of the wiper, the heater / A/C unit must be removed.

92 15 19 Removing and installing wiper motor (without A/C system)

Vehicles without air conditioning

Removal

1. Disconnect battery and cover battery or terminals.

Note

Follow safety precautions for work on vehicles with airbags.

2. Remove steering wheel and steering column panel.

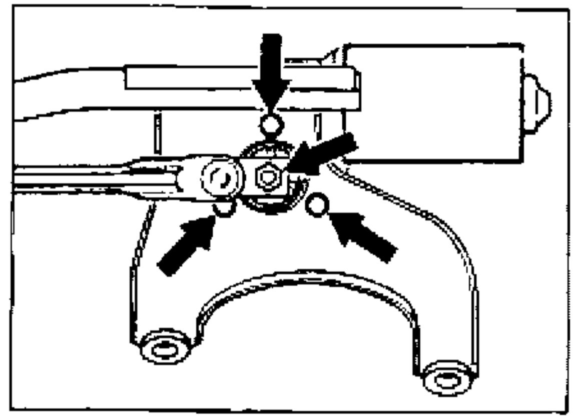
3. Remove tachometer and large and small instrument cluster.

4. Remove air guide. Disconnect six-pin connector from wiper motor.

5. Unhook left pneumatic spring of luggage compartment lid and support luggage compartment lid.

6. Remove cover of heater / A/C unit (old type with two screws and washers; new type with clamping spring).

7. Undo mounting nut of linkage, holding a 21 mm open-end wrench against linkage.



2072-92

Tightening torque: 22 ± 2 Nm (16 ± 1.5 ftlb.)

8. Release mounting screws. Tightening torque: 11 Nm (8 ftlb.).

9. Remove wiper motor through installation opening of tachometer.

Installation

Note

Installation of the wiper motor is much easier if the motor is fixed in position with a stud bolt (sawn-off screw) and a nut. After fixing the motor, fasten it in place with 2 screws. Then unscrew the nut and stud bolt and insert the third screw.

1. Connect battery and switch wiper motor on and off (park position).

2. Install linkage. Cable and linkage must form a straight line.

3. Switch on wiper motor and check wiper positions.
4. **Disconnect battery again** and install the parts that you have removed.
5. Replace the two self-locking screws of the airbag unit in the steering wheel.

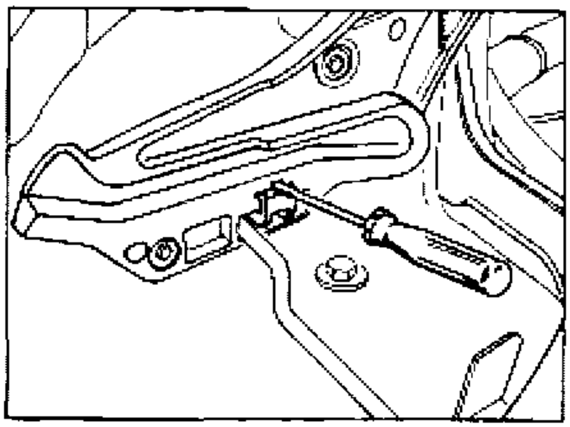
Note

To remove the crank mechanism of the wiper, the heater / A/C unit must be removed.

94 15 19 Removing and installing main headlight mountings

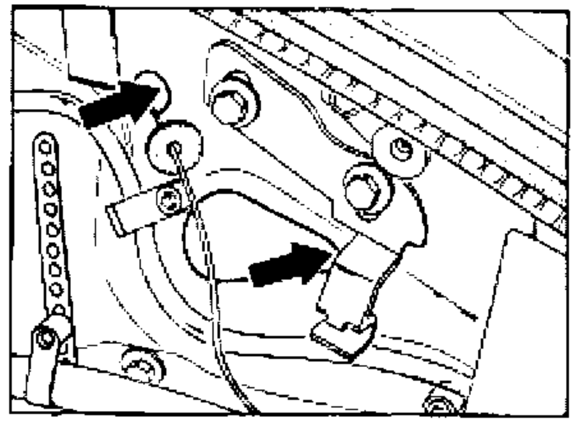
Removal

1. Unlock the main headlight in the luggage compartment and remove main headlight. Unclip safety clip in mounting using a screwdriver and remove lock lever with rubber sleeve.



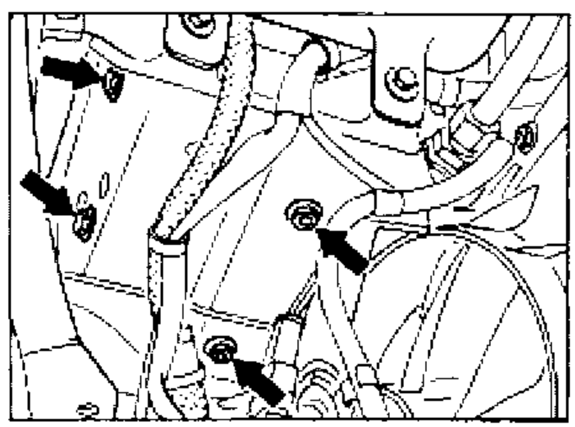
1885-94

2. **Only loosen** headlight mounting through openings in inside of luggage compartment using a 5 mm Allen socket wrench. Pull mounting up slightly and remove it.



1886-94

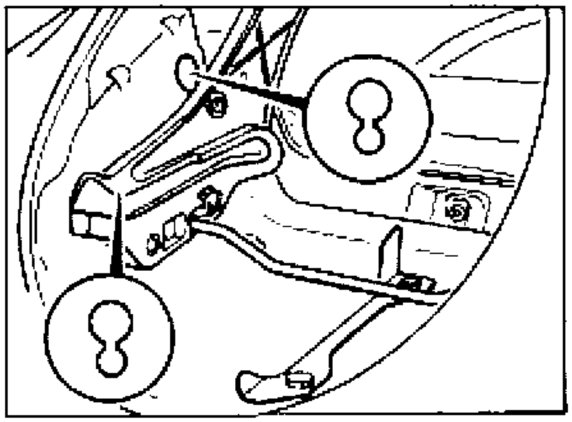
3. Remove lower part of front spoiler. **Only loosen** the M 6 hexagon head screws from below and take out the mounting. Unhook the electric connector. Loosen headlight rail from below and remove it.



1887-94

Installation

1. Position, but do not tighten, the headlight rail and mountings. Install and secure the locking lever.



1888-94

2. Install headlight and lock it in position with the locking lever. A second person must press the headlight firmly into the fender. Make sure that the headlight seal is correctly positioned in the fender.

3. Tighten the mountings in the correct order! The headlight rail must be tightened first, then the left and right mountings.
   Tightening torque: 10 Nm (7 ftlb.)

4. Check whether the headlight can be locked and unlocked. Adjust headlight using adjustment unit (see maintenance operations, page 03 - 20).

2. Flip over locking clamp on headlight hou ing and take off cover with fitted igniter. Screw out electrical connector from D2S lamp (quarter-turn socket fastener).

3. Carefully bend tab washers (2 ea.) open at retaining ring. Press retaining ring down and turn left (quarter-turn socket fastener). Take out retaining ring and D2S lamp.

2. Place headlight housing into fender a lock housing in place. Engage lockin

3. Check operation and adjustment of h lights (refer to maintenance operatior

Troubleshooting

Note

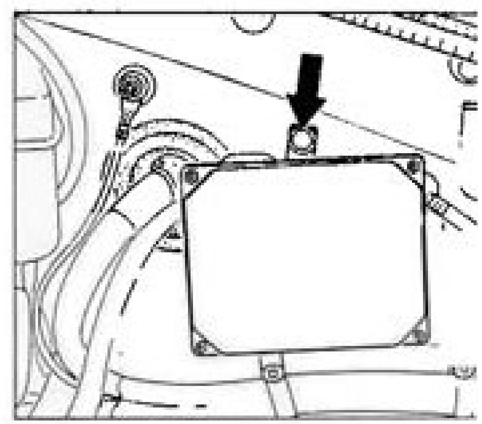
If the dipped beam (Litronic headlight) is defective, only the individual components can be replaced. The wires and control units are connected to **high voltage**. Do not use **any measuring device** to measure the voltage of wires or control units.

Removing and installing gas dischar lamp (D2S)

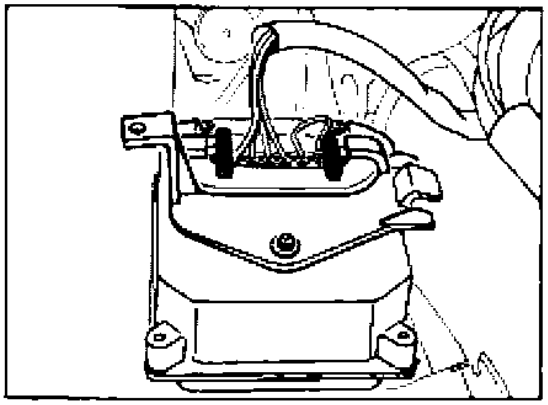
See page 94 - 3

Removing and installing control unit

1. Loosen fastening screw on control u bracket and remove control unit.



2. Disconnect electric plug connection and detach bracket from control unit.



2232-94

3. Install new control unit and check operation.

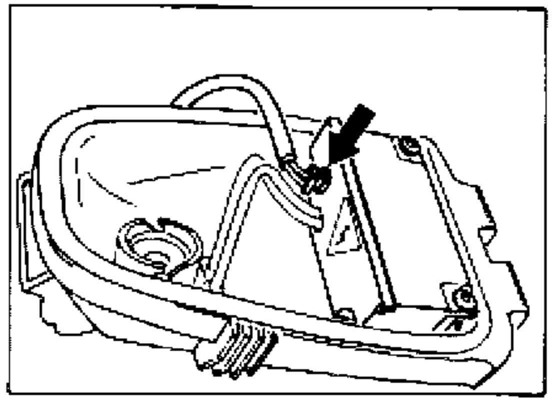
Removing and installing trigger device

Note

To check operation, the trigger unit of the second headlight can be installed.

1. Remove headlight housing. Fold back retaining bracket on headlight housing and remove lid with integrated trigger device.

2. Unscrew electric plug connection on D2S lamp (bajonet catch) and separate plug connection on trigger device.



2230-94

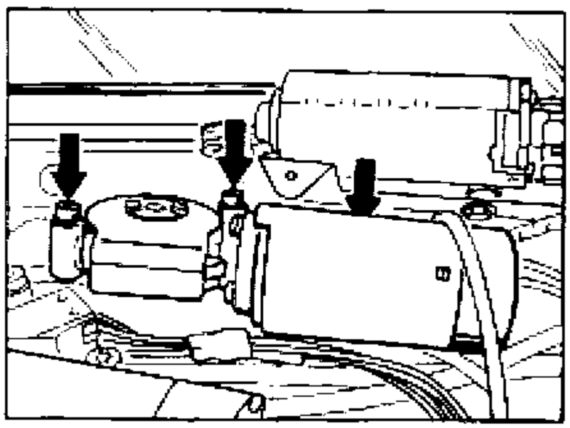
3. Install lid with integrated trigger device of second headlight and check operation.

92 00 19 Removing and installing rear window wiper system

Removal

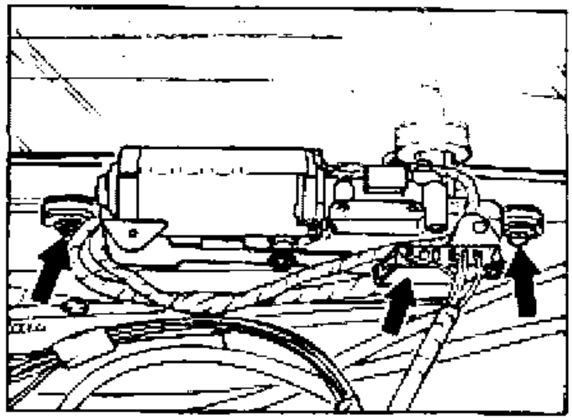
1. Disconnect and remove rear wiper arm. Take off cap and unscrew M 16 x 1 fastening nut. Remove plastic bushing. The rubber seal must remain in the rear window. **Do not** remove it.

2. Remove rear wall trim panel (4 screws) and cover for rear wiper motor (2 screws). Remove the sun blind motor in front of the wiper motor and place it on one side.



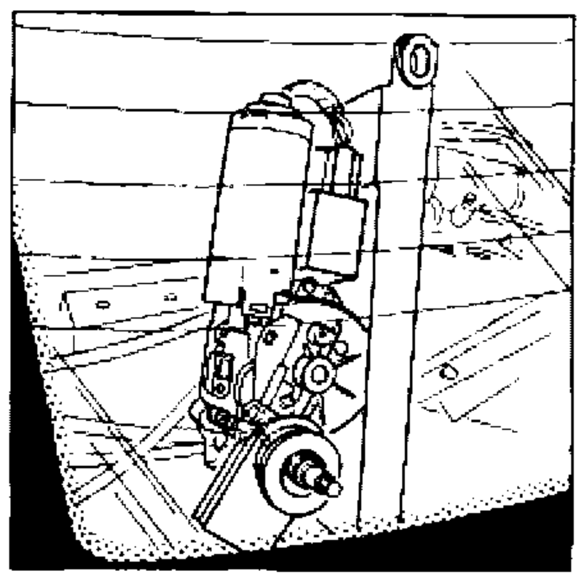
2251 - 92

3. Disconnect electrical connector from wiper motor. Unscrew and remove fastening screws (2 screws) of wiper system.



2252-92

4. Swing the wiper system up about 90 ° (the wiper motor points upwards) and carefully take it down out of the rear window (twisting it out).



2253-92

Installation

When installing the wiper system, take care not to damage the rear window pane or the rear window heater.

Torque specifications:

M 6 x 25 hexagonal head bolt	9 Nm (6.6 ftlb)
M 16 x 1 hexagonal nut	8...10 Nm (6...7.5 ftlb)
M 8 hexagonal nut	14...16 Nm (10.5...12 ftlb)